

MCA40140

**Data Communication and
Security**



Introduction

- A **Computer Network** is a set of Nodes connected by communication links

Node:

- Capable of sending/receiving data
- Examples: Computer, Mobile, Tablet, Printer, Camera, Switches, Bridge, Router etc.

Communication Link:

- Medium for sending/receiving data
- Two types of communication link: Wired and Wireless

Introduction

❖ Wired Communication Links

☐ Ethernet Cable

- 10 Base T, 100 Base T
- 10 Base Tx {RJ45 2 pair}

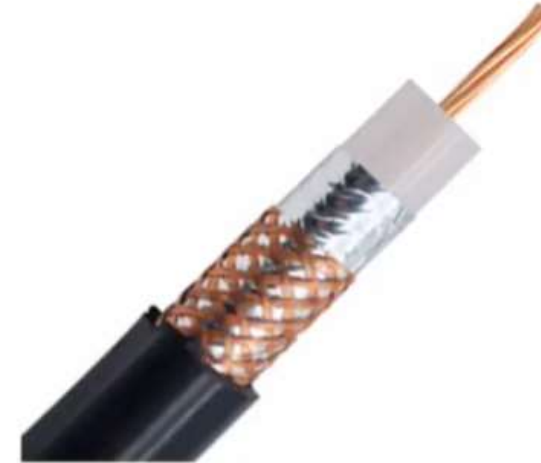
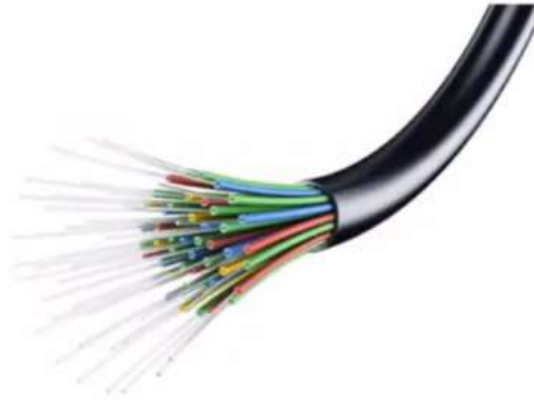
☐ Fiber optic Cable

- 100 Base Fx

☐ Coaxial Cable

- 10 Base 2, 100 Base 5

☐ USB Cable



Introduction

❖ Wireless Communication Links

☐ Infrared

- Used for short range {1-10m}
- Example: TV remote

☐ Radio

- Used for short range {1-100m}
- Example: Bluetooth, Wi-Fi, Zigbee

☐ Microwave

- Used for Medium range {1 - 10km}
- Example: Cellular System

☐ Satellite

- Used for Long range {above 100km}
- Example: Dish TV, GPS



Types of Node

❖ End Nodes of Computer Network

❑ Examples: Computer, Smart Phone, Tablet, Printer, Server, VoIP Phones, Cameras etc.

❖ Intermediary Nodes of Computer Network

❑ Examples: Routers, Switches, Bridges, Hubs, Repeaters, Cell Tower etc.

Introduction

Services of Computer Network

- ❖ e-Mail
- ❖ Internet {World Wide Web}
- ❖ Voice over IP
- ❖ Video Telephony
- ❖ File Sharing
- ❖ Storage Services
- ❖ Instant Messaging
- ❖ Online Gaming



Introduction

Types of Data Communication

❖ There are three types of data communication in computer network

- ☐ Simplex
- ☐ Half Duplex
- ☐ Full Duplex

❖ Simplex communication in computer network

- ☐ Simplex communication is unidirectional.
- ☐ One device will transmit and second will receive.
- ☐ Example : Radio station broadcasting

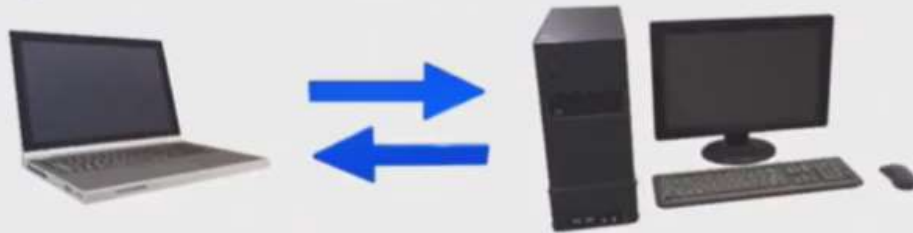


Introduction

Types of Data Communication

❖ Half Duplex communication in computer network

- ❑ Half Duplex communication is bidirectional.
- ❑ both device can send and receive data but not at same time.
- ❑ Example : Walkie Talkie



❖ Full Duplex communication in computer network

- ❑ Full Duplex communication is bidirectional.
- ❑ both device can send and receive data at same time
- ❑ Example : Telephony



Introduction

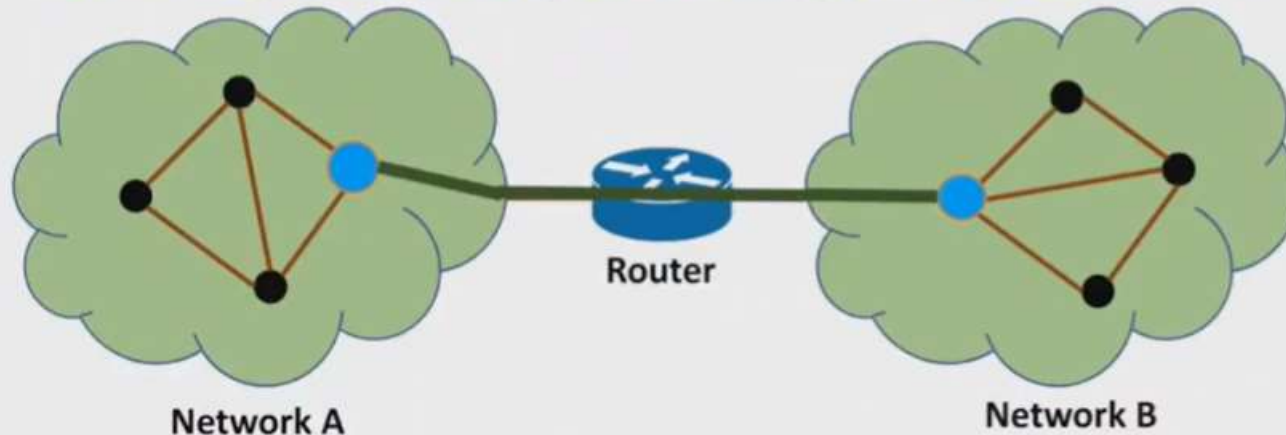
Types of Casting

❖ In general we use three types of casting in computer network

- ☐ Unicast
- ☐ Multicast
- ☐ Broadcast

❖ Unicast in computer network

- ☐ Unicast means one to one message forwarding.
- ☐ Using IP of node, unicasting can be done.



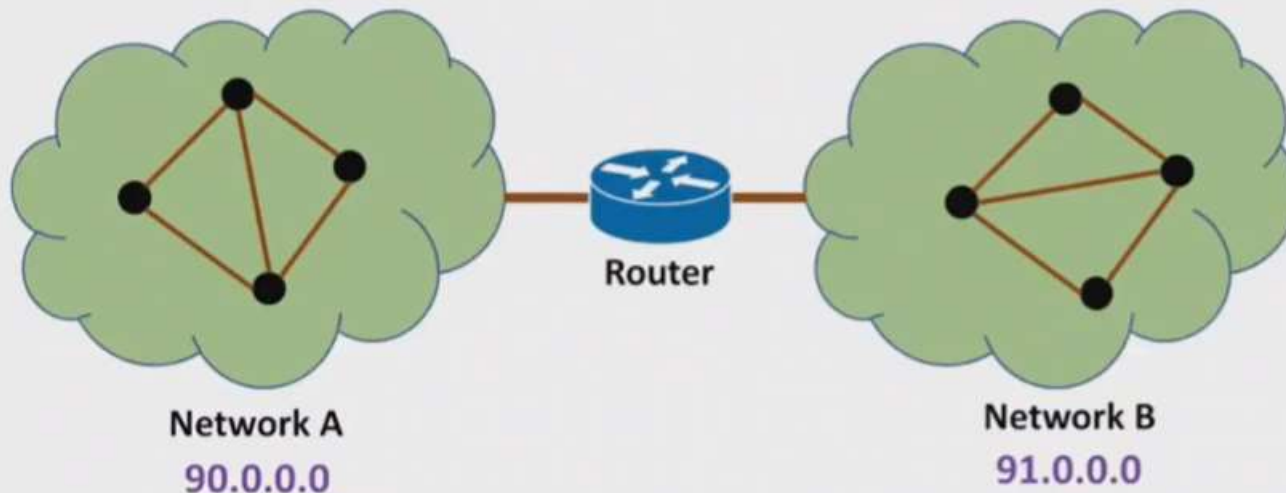
Introduction

Types of Casting

❖ Broadcast in computer network

□ There are two types of broadcast in computer network

1. **Limited Broadcast** : Limited Broadcast address is 255.255.255.255. It is limited to given router.
2. **Direct Broadcast** : Direct broadcast is used to send data to another network. It is not limited to given router.



Limited Broadcast in network A
by any node of network A is
done by 255.255.255.255

Direct Broadcast in network B
by any node of network A is
done by 91.255.255.255

Introduction

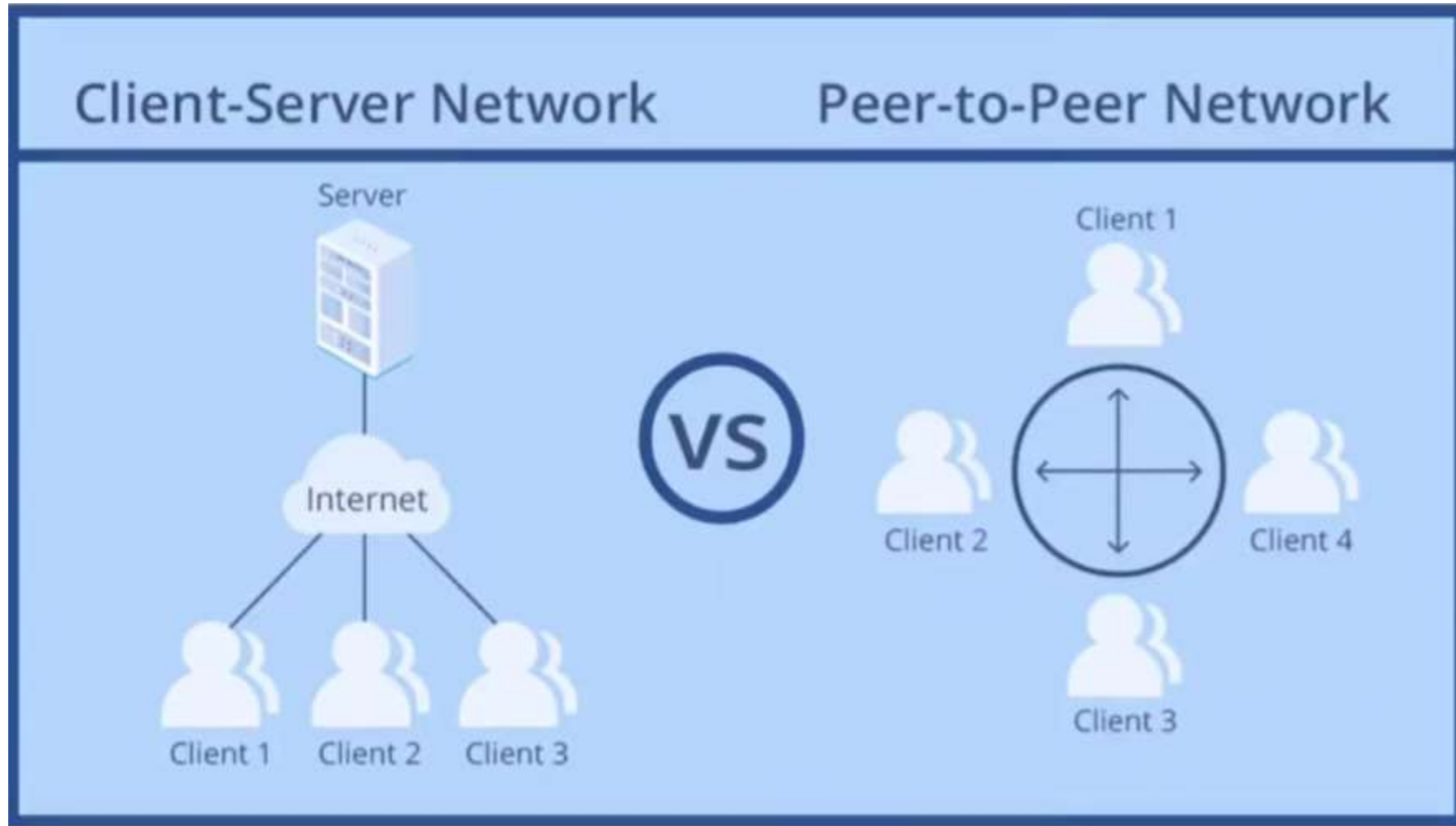
Types of Casting

❖ Multicast in computer network

❑ In multicast we send message from one computer many but not to all.

BROADCAST	MULTICAST
A method of transferring a message to all recipients simultaneously	A group communication where data transmission is addressed to a group of destination computers simultaneously
Packets are transmitted to all the connected devices in the network	Packets are transmitted to some of the devices in the network
There is no need for group management	Requires group management
Less secure	More secure
More traffic	Less traffic
Slower	Faster

Peer to Peer Network vs Client Server Network



Peer to Peer Network vs Client Server Network

PEER TO PEER NETWORK

A distributed application architecture that partitions tasks or workloads between peers

Each node can request for services and provide services

A decentralized network

CLIENT SERVER NETWORK

A distributed application structure based on resource or service providers called servers and service requesters called clients

Client requests for service and server responds with a service

A centralized network

PEER TO PEER NETWORK

Reliable as there are multiple service providing nodes

Service requesting node does not need to wait long

Expensive to implement

Comparatively less stable

CLIENT SERVER NETWORK

Clients depend on the server - failure in the server will disrupt the functioning of all clients

Access time for a service is higher

Does not require extensive hardware to set up the network

More stable and secure



Physical Structure of Computer Network

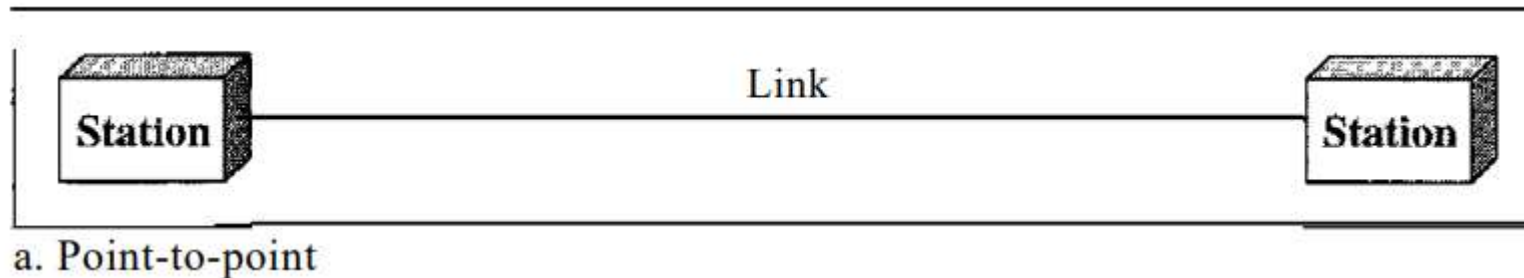
Type of Connection

- ❖ A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another.
- ❖ For visualization purposes, it is simplest to imagine any link as a line drawn between two points.
- ❖ For communication to occur, two devices must be connected in some way to the same link at the same time.
- ❖ There are **two possible types of connections**:
 - ✓ point-to-point and
 - ✓ multipoint.

Physical Structure of Computer Network

Point-to-Point

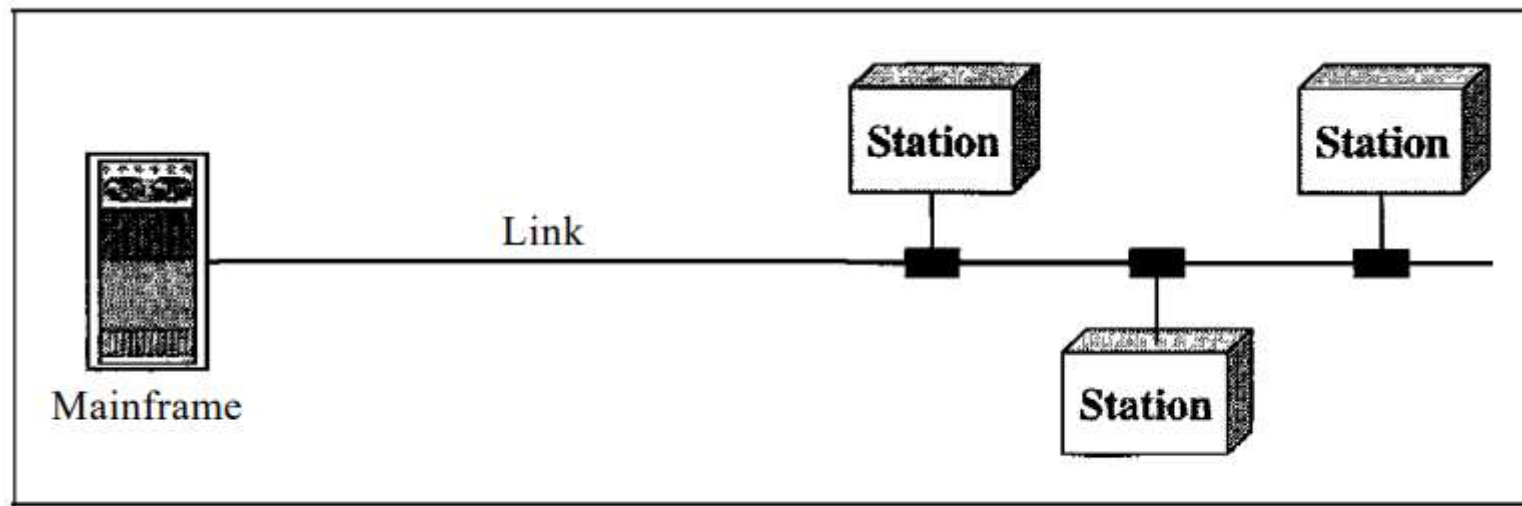
- ❖ A point-to-point connection provides a dedicated link between two devices.
- ❖ The entire capacity of the link is reserved for transmission between those two devices. Most point-to-point connections use an actual length of wire or cable to connect the two ends.



Physical Structure of Computer Network

Multipoint

- ❖ A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link.
- ❖ In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a spatially shared connection. If users must take turns, it is a timeshared connection.

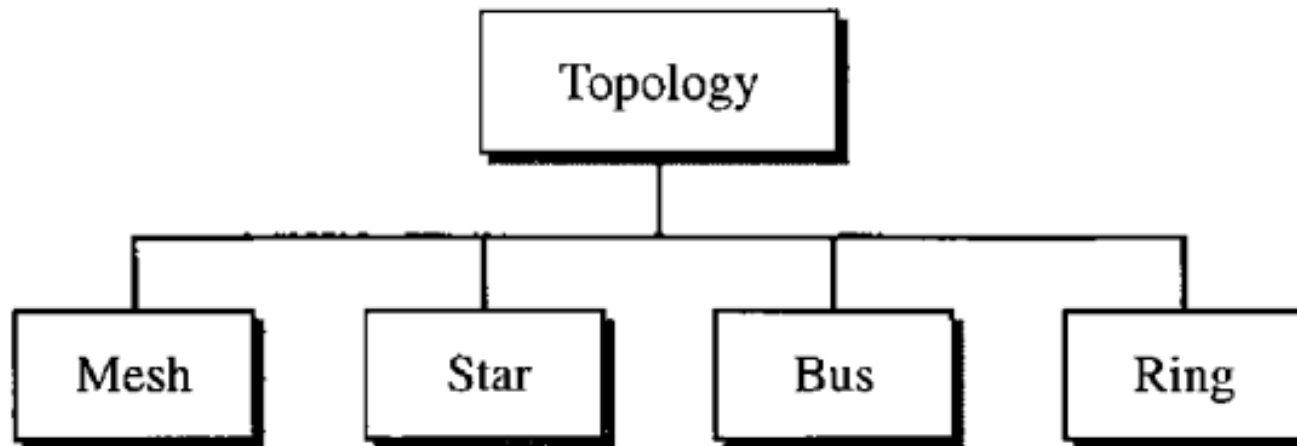


b. Multipoint

Physical Structure of Computer Network

Physical Topology

- ❖ refers to the way in which a network is laid out physically: Two or more devices connect to a link; two or more links form a topology.
- ❖ The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.





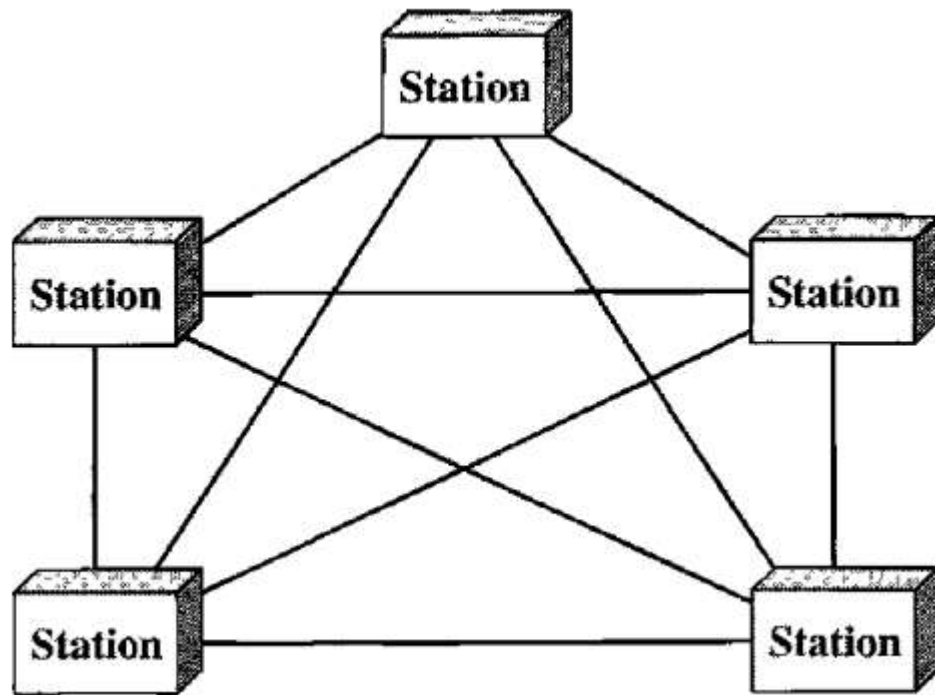
Physical Structure of Computer Network

Mesh Topology

- ❖ In a mesh topology, every device has a dedicated point-to-point link to every other device.
- ❖ The term dedicated means that the link carries traffic only between the two devices it connects.
- ❖ To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node.
- ❖ Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, and finally node n must be connected to $n - 1$ nodes.
- ❖ We need $n(n - 1)$ physical links.

Physical Structure of Computer Network

Mesh Topology



- ❖ However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2.
- ❖ In other words, we can say that in a mesh topology, we need
$$\frac{n(n-1)}{2}$$
 duplex-mode links.



Physical Structure of Computer Network

Mesh Topology

Advantages

- ❖ The use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems.
- ❖ mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.
- ❖ Privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.
- ❖ Finally, point-to-point links make fault identification and fault isolation easy. Traffic can be routed to avoid links with suspected problems. This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution



Physical Structure of Computer Network

Mesh Topology

Disadvantages

- ❖ Every device must be connected to every other device, installation and reconnection are difficult.
- ❖ The sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.
- ❖ The hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

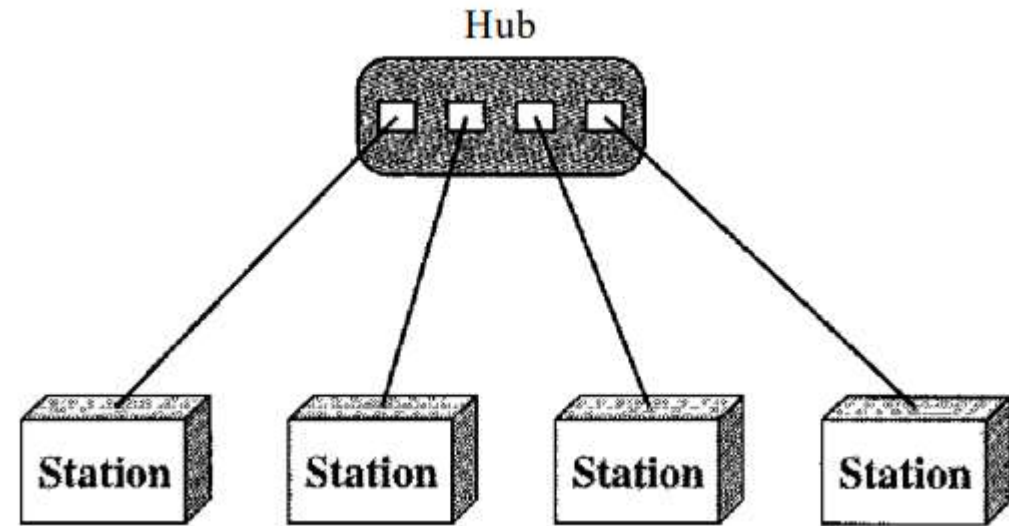
Practical example:

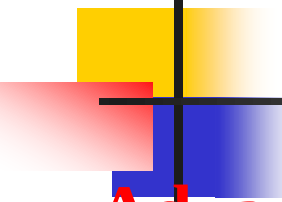
- ❖ The connection of telephone regional offices in which each regional office needs to be connected to every other regional office.

Physical Structure of Computer Network

Star Topology

- ❖ In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub.
- ❖ The devices are not directly linked to one another.
- ❖ Unlike a mesh topology, a star topology does not allow direct traffic between devices.
- ❖ The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.





Physical Structure of Computer Network

Advantages:

- ❖ A star topology is less expensive than a mesh topology. In a star, each device needs only one link and one I/O port to connect it to any number of others. This factor also makes it easy to install and reconfigure.
- ❖ Far less cabling needs to be housed, and additions, moves, and deletions involve only one connection: between that device and the hub.
- ❖ Star topology is robust. If one link fails, only that link is affected. All other links remain active. This factor also lends itself to easy fault identification and fault isolation.

Disadvantage:

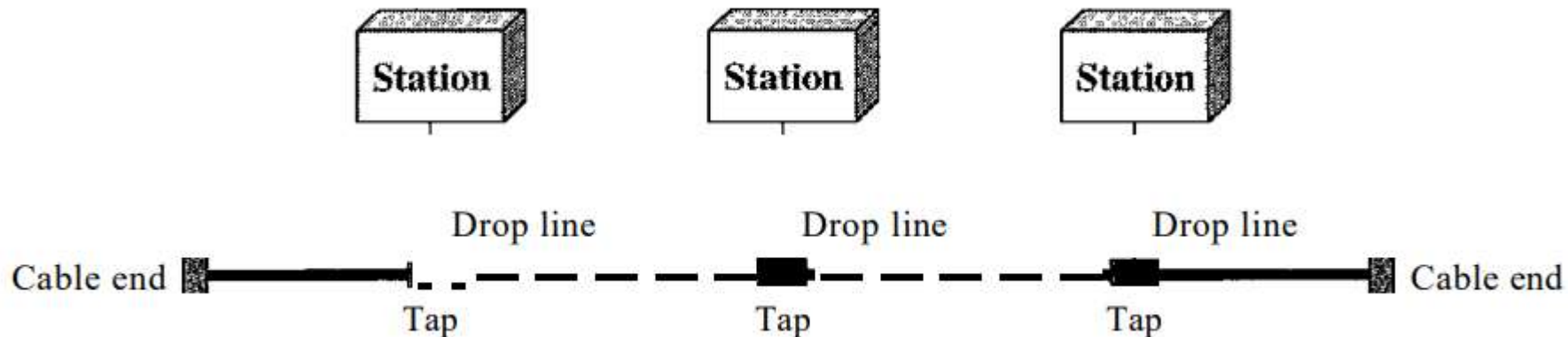
- ❖ dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.

Practical Example: Used in local-area networks (LANs)

Physical Structure of Computer Network

Bus Topology

- ❖ The preceding examples all describe point-to-point connections. A bus topology, on the other hand, is multipoint.
- ❖ One long cable acts as a backbone to link all the devices in a network
- ❖ Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.





Physical Structure of Computer Network

- ❖ As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason, there is a limit on the number of taps a bus can support and on the distance between those taps.

Advantages

- ❖ Ease of installation.
- ❖ Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies. In a star, for example, four network devices in the same room require four lengths of cable reaching all the way to the hub. In a bus, this redundancy is eliminated.
- ❖ Only the backbone cable stretches through the entire facility. Each drop line has to reach only as far as the nearest point on the backbone.



Physical Structure of Computer Network

Disadvantages

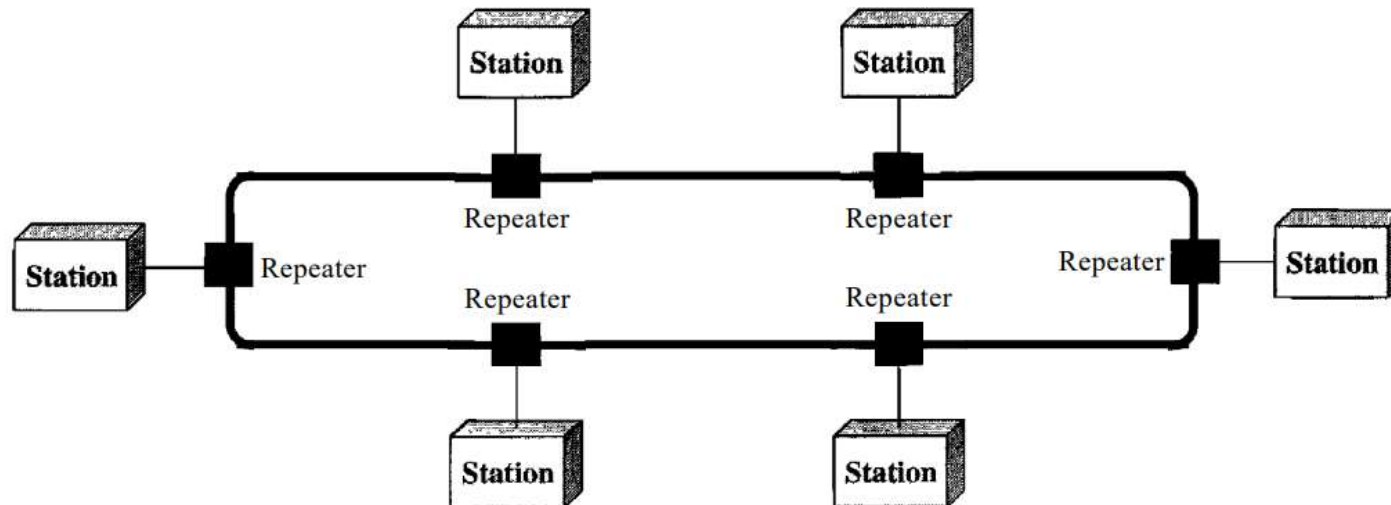
- ❖ Difficult reconnection and fault isolation.
- ❖ A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices.
- ❖ Signal reflection at the taps can cause degradation in quality. This degradation can be controlled by limiting the number and spacing of devices connected to a given length of cable. Adding new devices may therefore require modification or replacement of the backbone.
- ❖ In addition, a fault or break in the bus cable stops all transmission, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

Practical Example: Earlier used in the design of early local area networks. Ethernet LANs can use a bus topology, but they are less popular now.

Physical Structure of Computer Network

Ring Topology

- ❖ In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- ❖ A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- ❖ Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along





Physical Structure of Computer Network

Advantages:

- ❖ A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically).
- ❖ To add or delete a device requires changing only two connections. The only constraints are media and traffic considerations (maximum ring length and number of devices).
- ❖ In addition, fault isolation is simplified. Generally, in a ring, a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.



Physical Structure of Computer Network

Disadvantages:

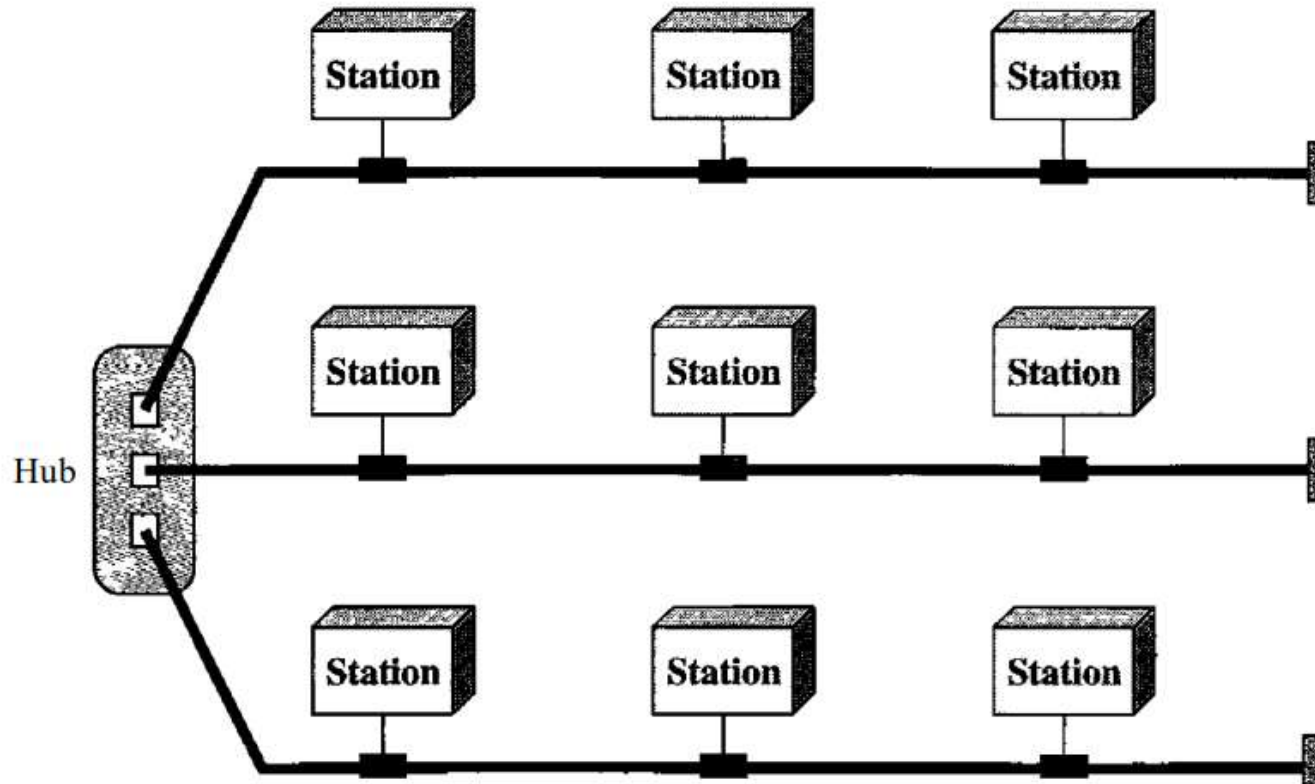
- ❖ In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break.

Practical Example: Ring topology was prevalent when IBM introduced its local-area network Token Ring. Today, the need for higher-speed LANs has made this topology less popular.

Physical Structure of Computer Network

Hybrid Topology

A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology





Types of Computer Network

- ❖ Local Area Network (LAN)
- ❖ Metropolitan Area Network (MAN)
- ❖ Wide Area Network (WAN)

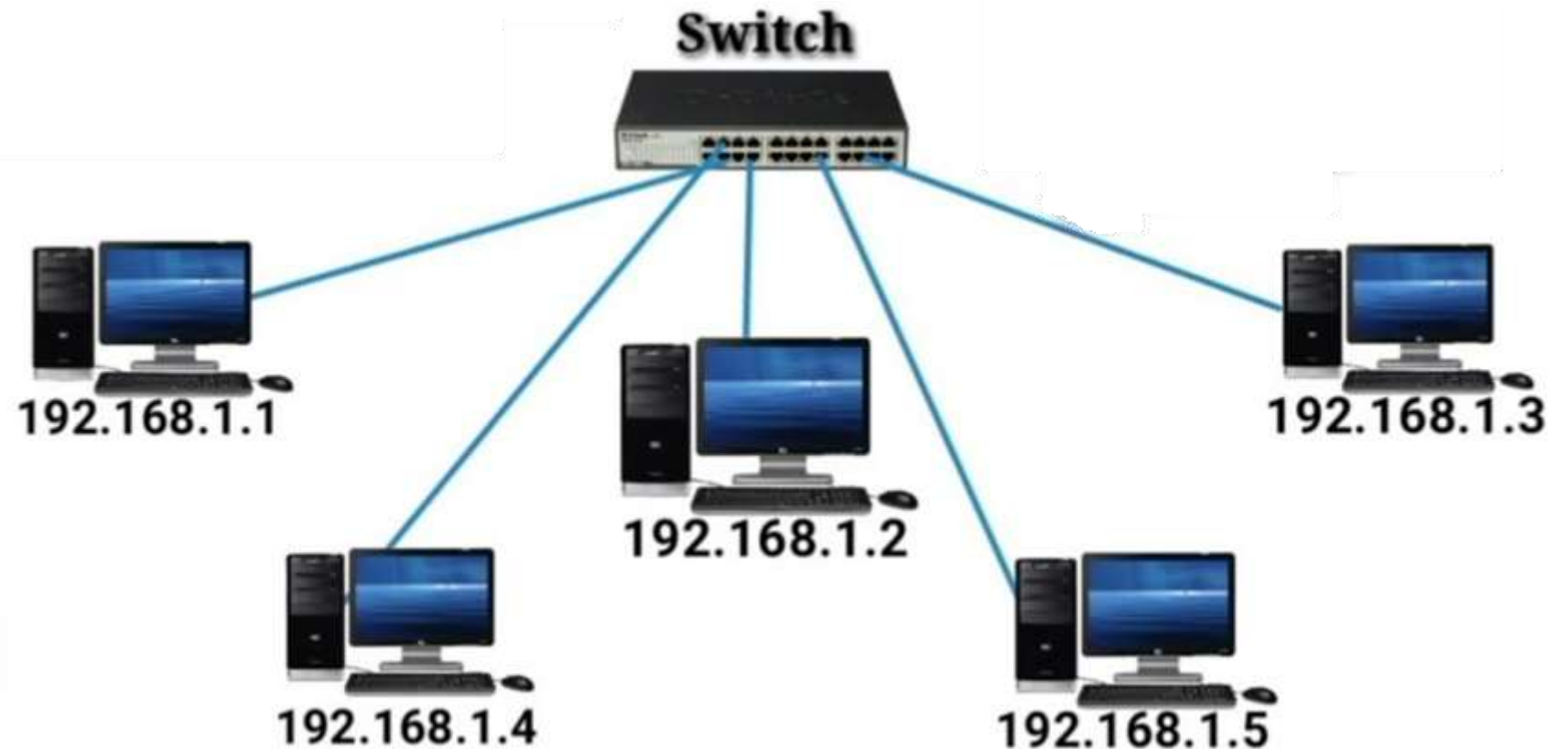
Local Area Network (LAN):

- ❖ LAN is a computer network that interconnects the computers within the limited area such as
 - residence
 - school
 - laboratory
 - University campus
 - Office building

Types of Computer Network

LAN - Devices

- ❖ Wired LAN (Example: Ethernet – Hub, Switch)
- ❖ Wireless LAN (Example: WiFi)





Types of Computer Network

Metropolitan Area Network (MAN):

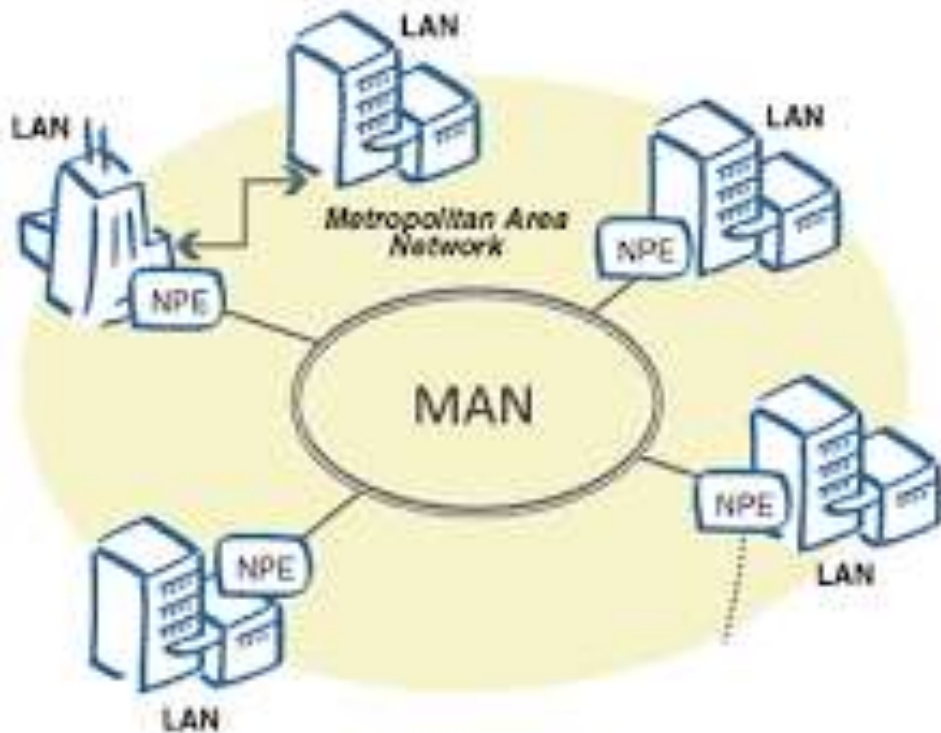
- ❖ A MAN is a network that interconnects users with computer resources in a geographic area or region larger than that covered by even a large local area network (but smaller than the area covered by a wide area network – metropolitan city)

MAN – Devices:

- ❖ Hub/Switch
- ❖ Routers/Bridges

Types of Computer Network

Metropolitan Area Network (MAN):



Example:

- ❖ Telephone Network
- ❖ Cable TV



Types of Computer Network

Wide Area Network (WAN):

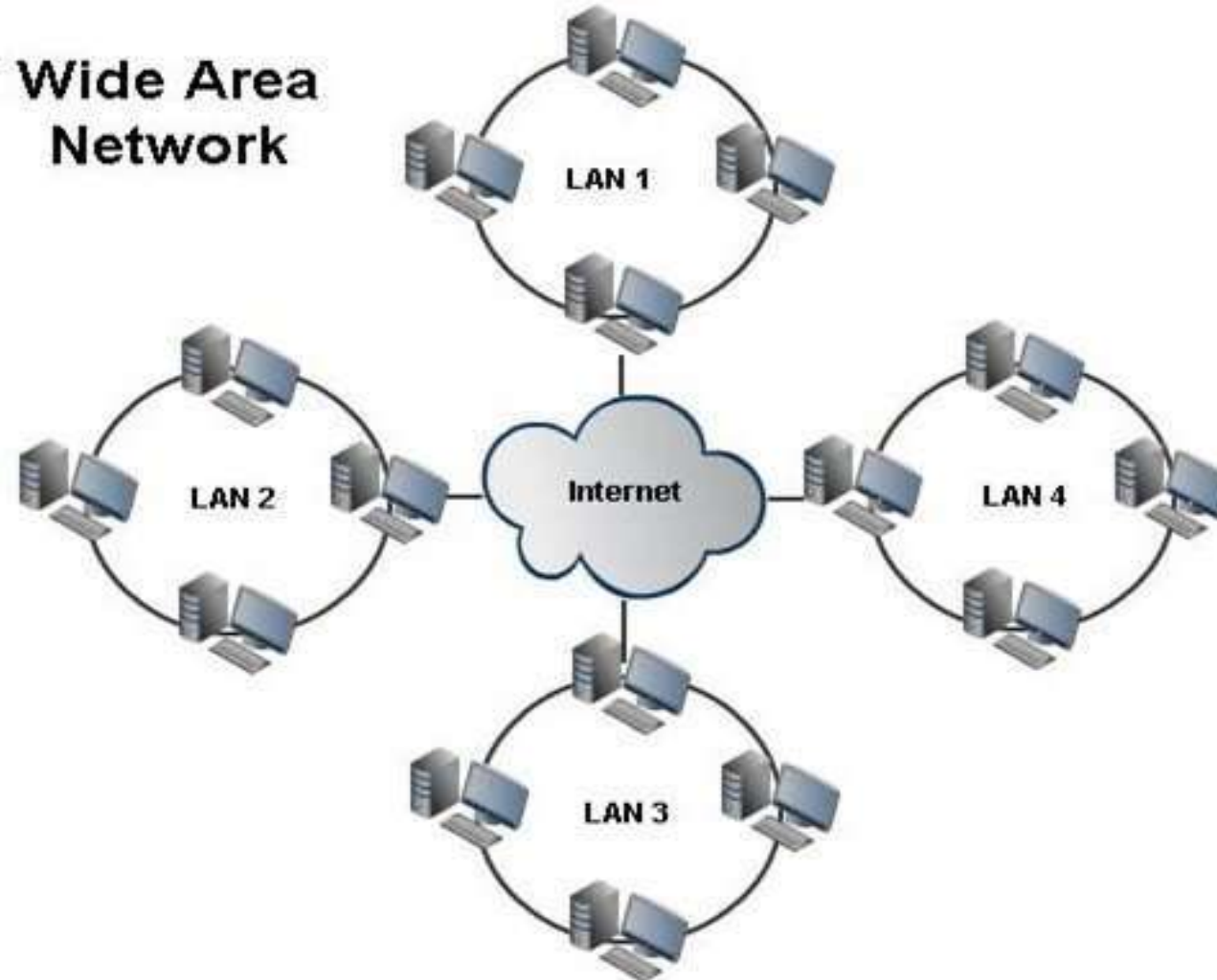
- ❖ WAN Provide a Long-Distance Transmission of Data (large geographical area)
- ❖ By Using WAN Exchange, the Information from one country to another country

WAN – Devices:

- ❖ End and intermediary devices

Types of Computer Network

Wide Area Network (WAN):





Types of Computer Network

Parameters	LAN	MAN	WAN
Full Form	Local Area Network	Metropolitan Area Network	Wide Area Network
Range	Up to 2Km	5-50Km	Above 50Km
Transmission Speed	Very High	Average	Low
Area	Building, Office	Within City	Within Countries
Ownership	Private	Private or Public	Private or Public
Maintenance	Easy	Difficult	Very Difficult
Error Rate	Low	High	Very High
Cost	Low	High	Very High



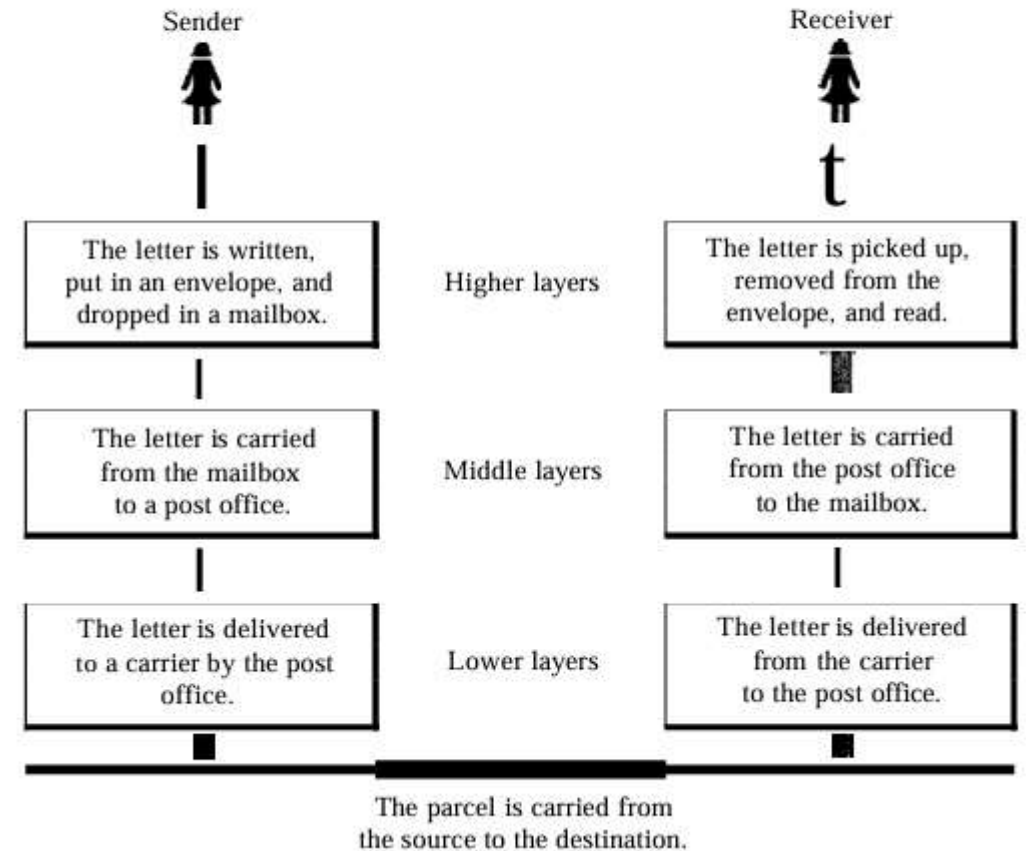
Network Models

- ❖ A network is a combination of hardware and software that sends data from one location to another.
- ❖ The hardware consists of the physical equipment that carries signals from one point of the network to another.
- ❖ The software consists of instruction sets that enable the functions or services in a computer network.
- ❖ A service is a set of operations that a layer provide. The service defines what operations the layer is prepared to perform on behalf of its users.
- ❖ Standards are needed so that these heterogeneous networks can communicate with one another.
- ❖ The two best-known standards are the **OSI model** and the **Internet model**.
- ❖ The OSI (Open Systems Interconnection) model defines a **seven-layer network**; the Internet model defines a **five-layer network**.

Network Models

- ❖ A layered approach in computer networks decomposes a problem of building a computer network into manageable task.
- ❖ Layered approach allow the different services to be performed in each layer.
- ❖ Each layer at the sending site uses the services of the layer immediately below it.

Tasks involved in sending a letter





Network Models

Layered Architecture:

OSI Model

- ❖ The layered model that dominated data communications and networking literature before 1990 was the Open Systems Interconnection (OSI) model.
- ❖ It is a model for understanding and designing a network architecture that is flexible, robust and interoperable.
- ❖ Developed by ISO
- ❖ It is only a guideline and hence it is referred as OSI reference model.
- ❖ The purpose of the OSI model is to show how to facilitate the communication between different systems without requiring changes to the logic of the underlying hardware and software
- ❖ The OSI model was never fully implemented.



Network Models

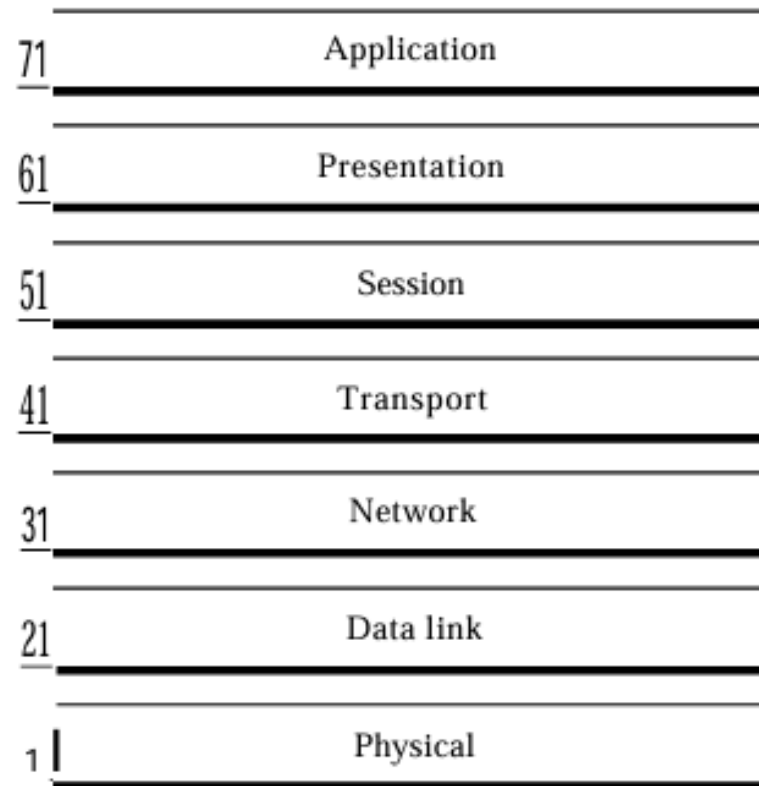
Layered Architecture:

TCP/IP

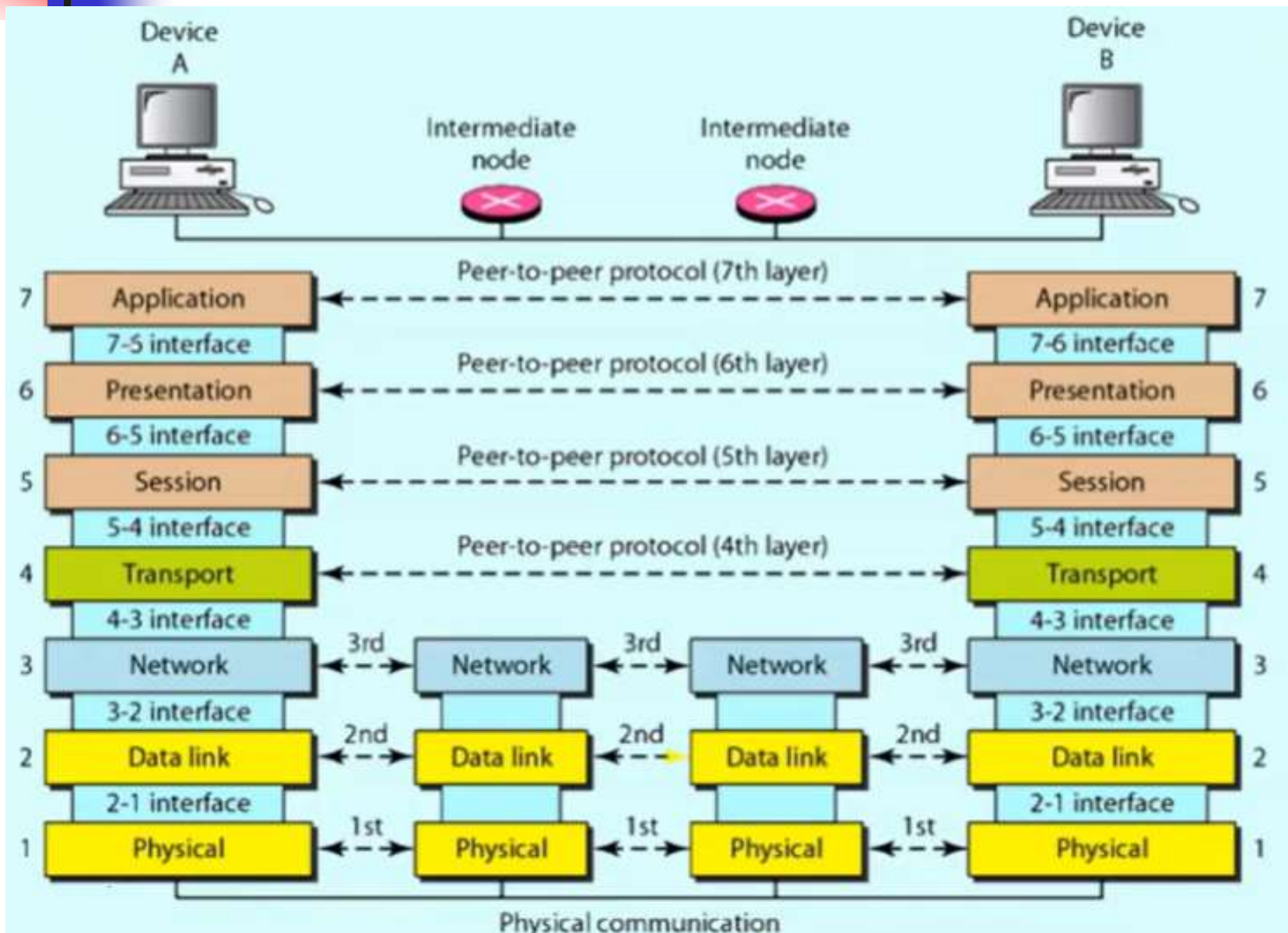
- ❖ The TCP/IP protocol suite became the dominant commercial architecture because it was used and tested extensively in the Internet.
- ❖ TCP/IP is a hierarchical protocol made up of interactive modules each of which provides specific functionality.

OSI Reference Model

- ❖ **Purpose:** The OSI model is to facilitate the communication between different system without requiring changes to the logic of the underlying hardware and software

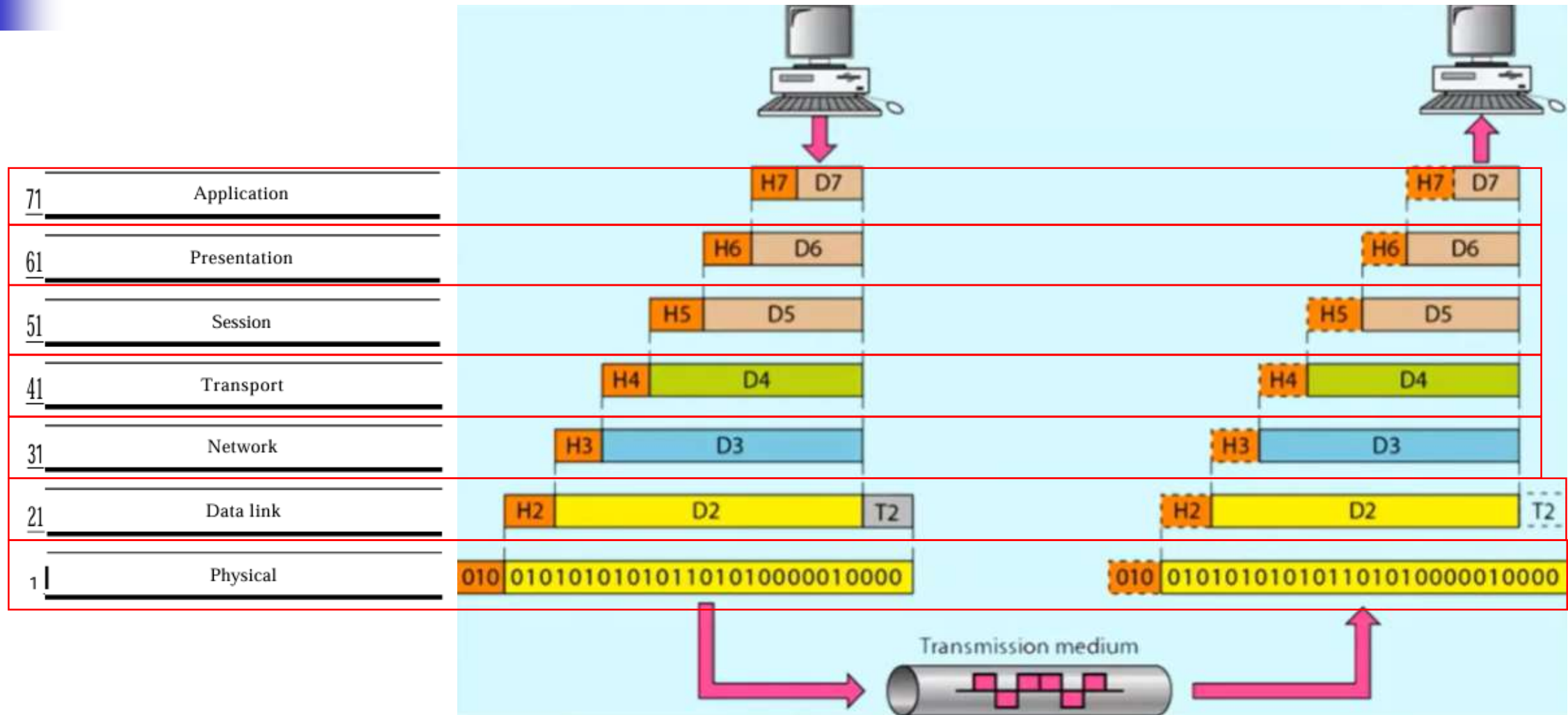


OSI Reference Model



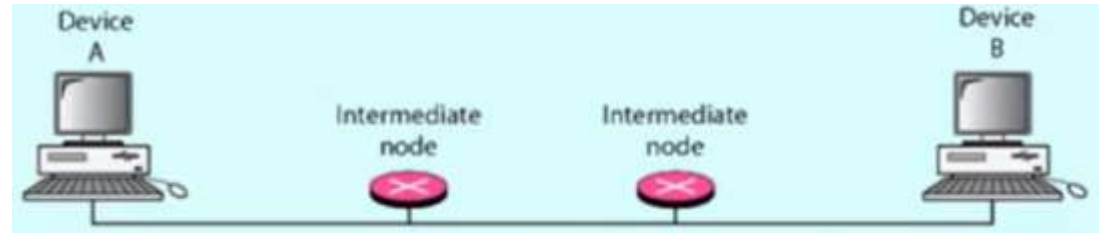
- ❖ Router is Layer 3 device.
- ❖ Switches can be of Layer 2 or Layer 3.
- ❖ Bridge is Layer 2 device.
- ❖ Hub is Layer 1 device.
- ❖ Repeater is Layer 1 device.

OSI Reference Model



Encapsulation: A packet (header and data) at level 7 is encapsulated in a packet at level 6. The whole packet at level 6 is encapsulated in a packet at level 5, and so on.

OSI Reference Model



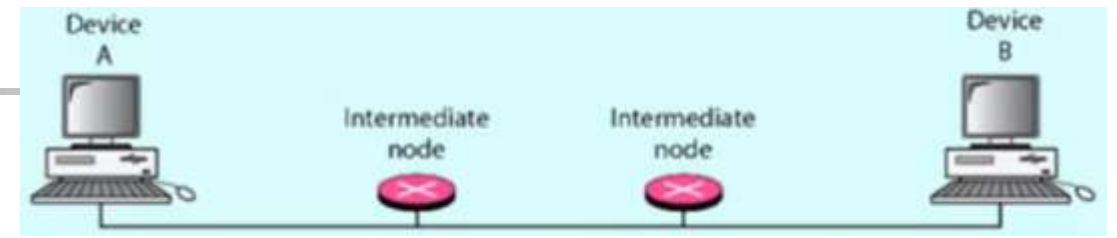
Application Layer

- ❖ The application layer enables the user to access the network.
- ❖ Services provided by the application layer:
 - ✓ **File transfer and access management (FTAM):** This application allows a user to access files in a remote host (to make changes or read data), to retrieve files from a remote computer for use in the local computer, and to manage or control files in a remote computer locally.
 - ✓ **Mail service:** This application provides the basis for e-mail forwarding and storage.
 - ✓ **Directory services:** This application provides distributed database sources and access for global information about various objects and services.

OSI Reference Model

Presentation Layer

- ❖ The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems.
- ❖ Services provided by the presentation layer:
 - ✓ **Translation:** The processes (running programs) in two systems are usually exchanging information in the form of character strings, numbers, and so on. The information must be changed to bit streams before being transmitted. Because different computers use different encoding systems, the presentation layer is responsible for interoperability between these different encoding methods. The presentation layer at the sender changes the information from its sender-dependent format into a common format. The presentation layer at the receiving machine, changes the common format into its receiver-dependent format.

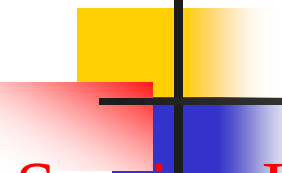




OSI Reference Model

Presentation Layer

- ❖ Services provided by the presentation layer:
 - ✓ **Encryption:** To carry sensitive information, a system must be able to ensure privacy. Encryption means that the sender transforms the original information to another form and sends the resulting message out over the network. Decryption reverses the original process to transform the message back to its original form.
 - ✓ **Compression:** Data compression reduces the number of bits contained in the information. Data compression becomes particularly important in the transmission of multimedia such as text, audio, and video.



OSI Reference Model

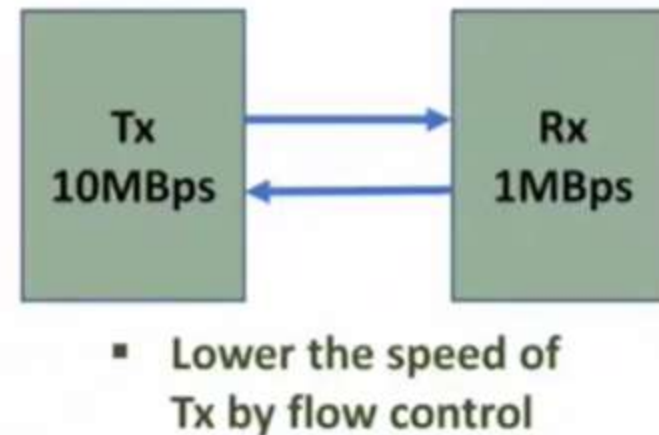
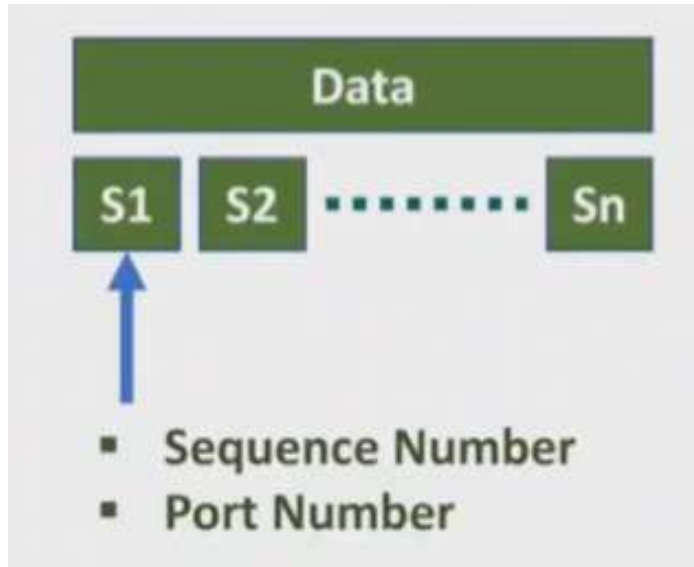
Session Layer

- ❖ It establishes, maintains, and synchronizes the interaction among communicating systems.
- ❖ Services provided by the session layer:
 - ✓ **Dialog control:** The session layer allows two systems to enter into a dialog. It allows the communication between two processes to take place in either half duplex (one way at a time) or full-duplex (two ways at a time) mode.
 - ✓ **Synchronization:** The session layer allows a process to add checkpoints, or synchronization points, to a stream of data. For example, if a system is sending a file of 2000 pages, it is advisable to insert checkpoints after every 100 pages to ensure that each 100-page unit is received and acknowledged independently. In this case, if a crash happens during the transmission of page 523, the only pages that need to be resent after system recovery are pages 501 to 523. Pages previous to 501 need not be resent.

OSI Reference Model

Transport Layer

- ❖ The transport layer is responsible for process-to-process delivery of the entire message. A process is an application program running on a host.
- ❖ The transport layer, on the other hand, ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level.





OSI Reference Model

Transport Layer

- ❖ Services provided by the transport layer:
 - ✓ **Segmentation and reassembly:** A message is divided into transmittable segments, with each segment containing a sequence number. These numbers enable the transport layer to reassemble the message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.
 - ✓ **Connection control:** The transport layer can be either connectionless or connection oriented. A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine. A connection-oriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets. After all the data are transferred, the connection is terminated.



OSI Reference Model

Transport Layer

- ❖ Services provided by the transport layer:
 - ✓ **Flow control**
 - ✓ **Error control:** Like the data link layer, the transport layer is responsible for error control. However, error control at this layer is performed process-to-process. The sending transport layer makes sure that the entire message arrives at the receiving transport layer without error (damage, loss, or duplication). Error correction is usually achieved through retransmission.
 - ✓ **Service-point addressing:** Computers often run several programs at the same time. For this reason, source-to-destination delivery means delivery not only from one computer to the next but also from a specific process (running program) on one computer to a specific process (running program) on the other. The transport layer header must therefore include a type of address called a service-point address (or port address).

OSI Reference Model

Network Layer

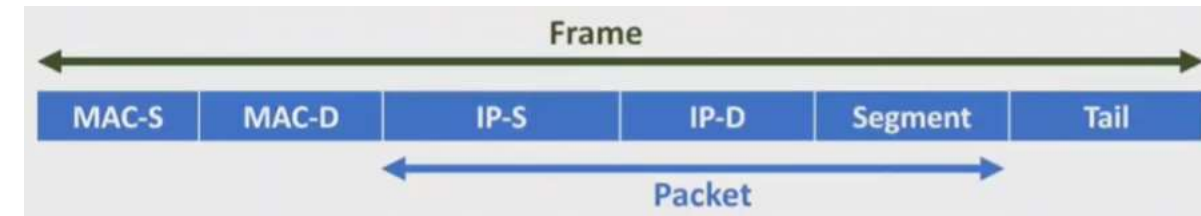
- ❖ The network layer is responsible for the delivery of a packet from source-to-destination network.
- ❖ Packet = segment + source and destination IP address (logical address – helps to find the network)
- ❖ Services provided by the network layer:
 - ✓ Logical addressing
 - ✓ Routing



OSI Reference Model

Data Link Layer

- ❖ The data link layer is responsible for moving frames from one hop (node) to the next.
- ❖ Frame = packet + MAC address of source & destination + tail (error detection)



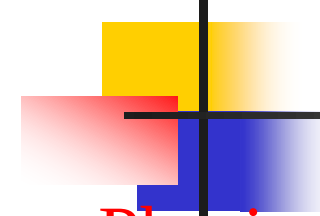
- ❖ Services provided by the data link layer:
 - ✓ **Framing:** The data link layer divides the stream of bits received from the network layer into manageable data units called frames.
 - ✓ **Physical addressing:** If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame. If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects the network to the next one.



OSI Reference Model

Data Link Layer

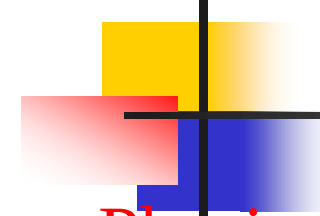
- ❖ Services provided by the data link layer:
 - ✓ **Flow control:** If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.
 - ✓ **Error control:** The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplicate frames. Error control is normally achieved through a trailer added to the end of the frame.
 - ✓ **Access control:** When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.



OSI Reference Model

Physical Layer

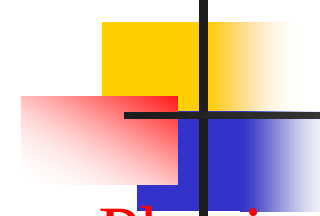
- ❖ The physical layer coordinates the functions required to carry a bit stream over a physical medium.
- ❖ It deals with the mechanical and electrical specifications of the interface and transmission medium.
- ❖ Services provided by the physical layer:
 - ✓ **Physical characteristics of interfaces and medium:** The physical layer defines the characteristics of the interface between the devices and the transmission medium. It also defines the type of transmission medium.
 - ✓ **Representation of bits:** The physical layer data consists of a stream of bits (sequence of 0's or 1's) with no interpretation. Bits must be encoded into signals--electrical or optical. The physical layer defines the type of encoding (how 0's and 1's are changed to signals).



OSI Reference Model

Physical Layer

- ❖ Services provided by the physical layer:
 - ✓ **Data rate:** The transmission rate-the number of bits sent each second-is also defined by the physical layer.
 - ✓ **Synchronization of bits:** The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level. In other words, the sender and the receiver clocks must be synchronized.
 - ✓ **Line configuration:** The physical layer is concerned with the connection of devices to the media. In a point-to-point configuration, two devices are connected through a dedicated link. In a multipoint configuration, a link is shared among several devices.

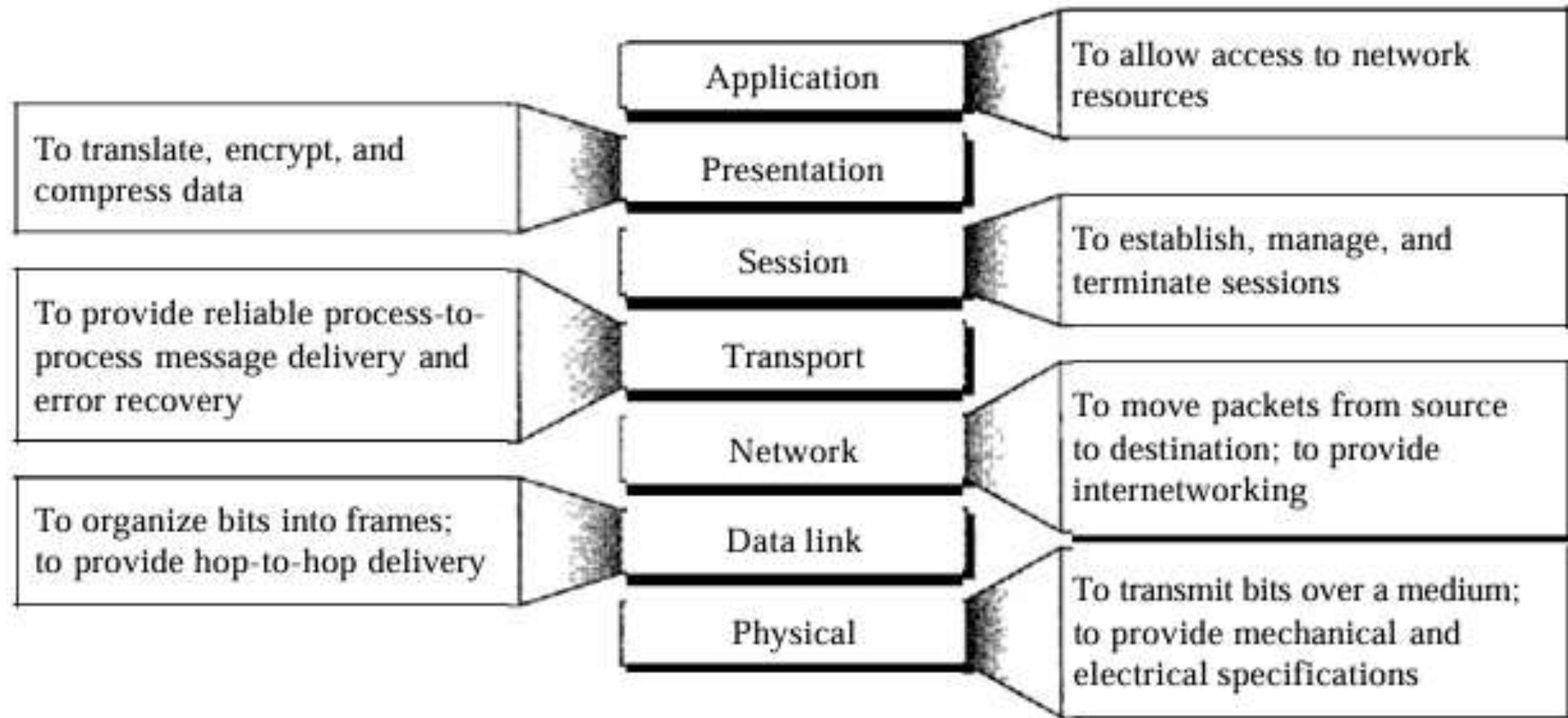


OSI Reference Model

Physical Layer

- ❖ Services provided by the physical layer:
 - ✓ **Physical topology:** The physical topology defines how devices are connected to make a network. Devices can be connected by using a mesh topology (every device is connected to every other device), a star topology (devices are connected through a central device), a ring topology (each device is connected to the next, forming a ring), a bus topology (every device is on a common link), or a hybrid topology (this is a combination of two or more topologies).
 - ✓ **Transmission mode:** The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex.

OSI Reference Model





Protocol Data Unit (PDU)

Application Layer – Data

Transport Layer – Segment

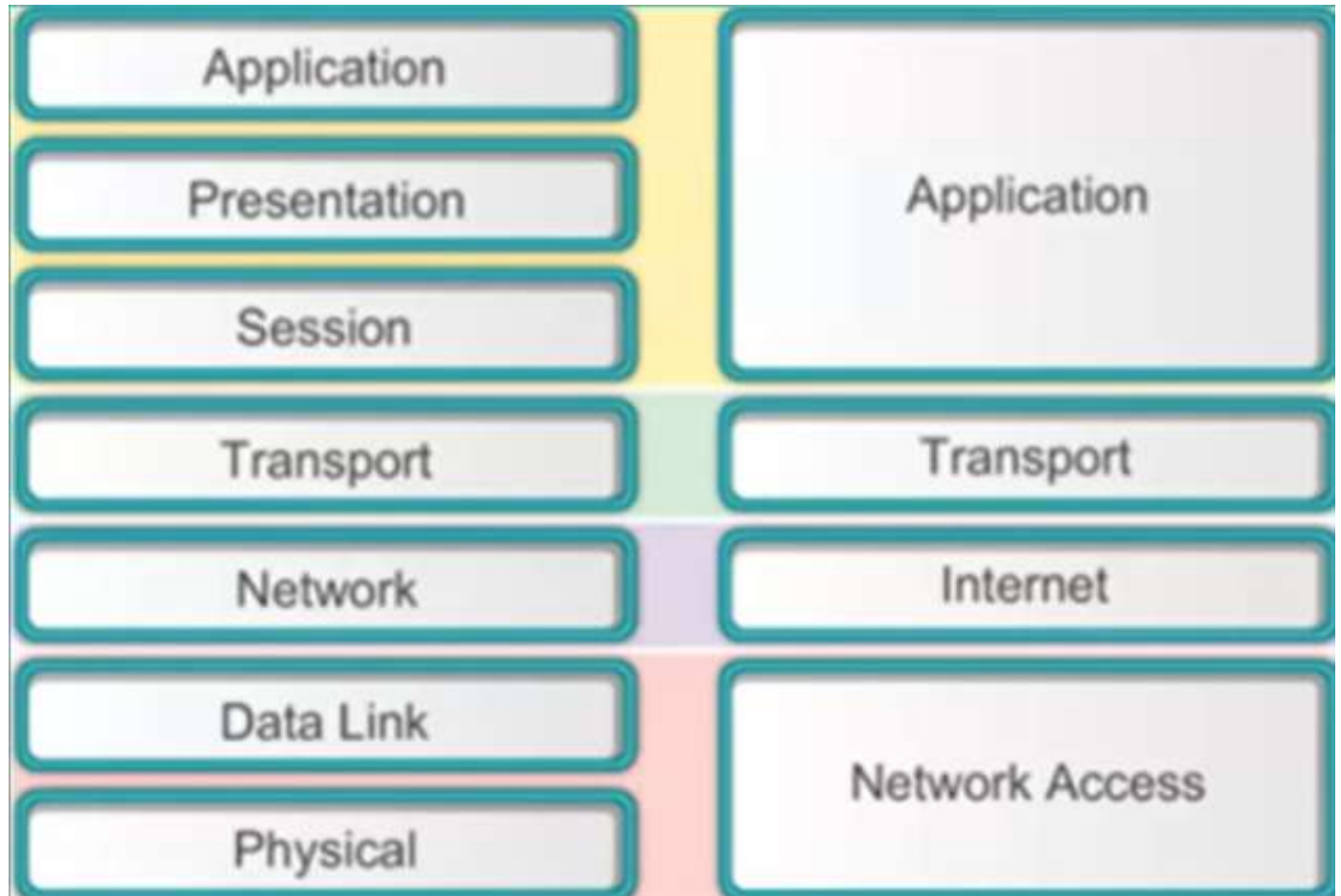
Network Layer – Packet

Data Link Layer – Frame

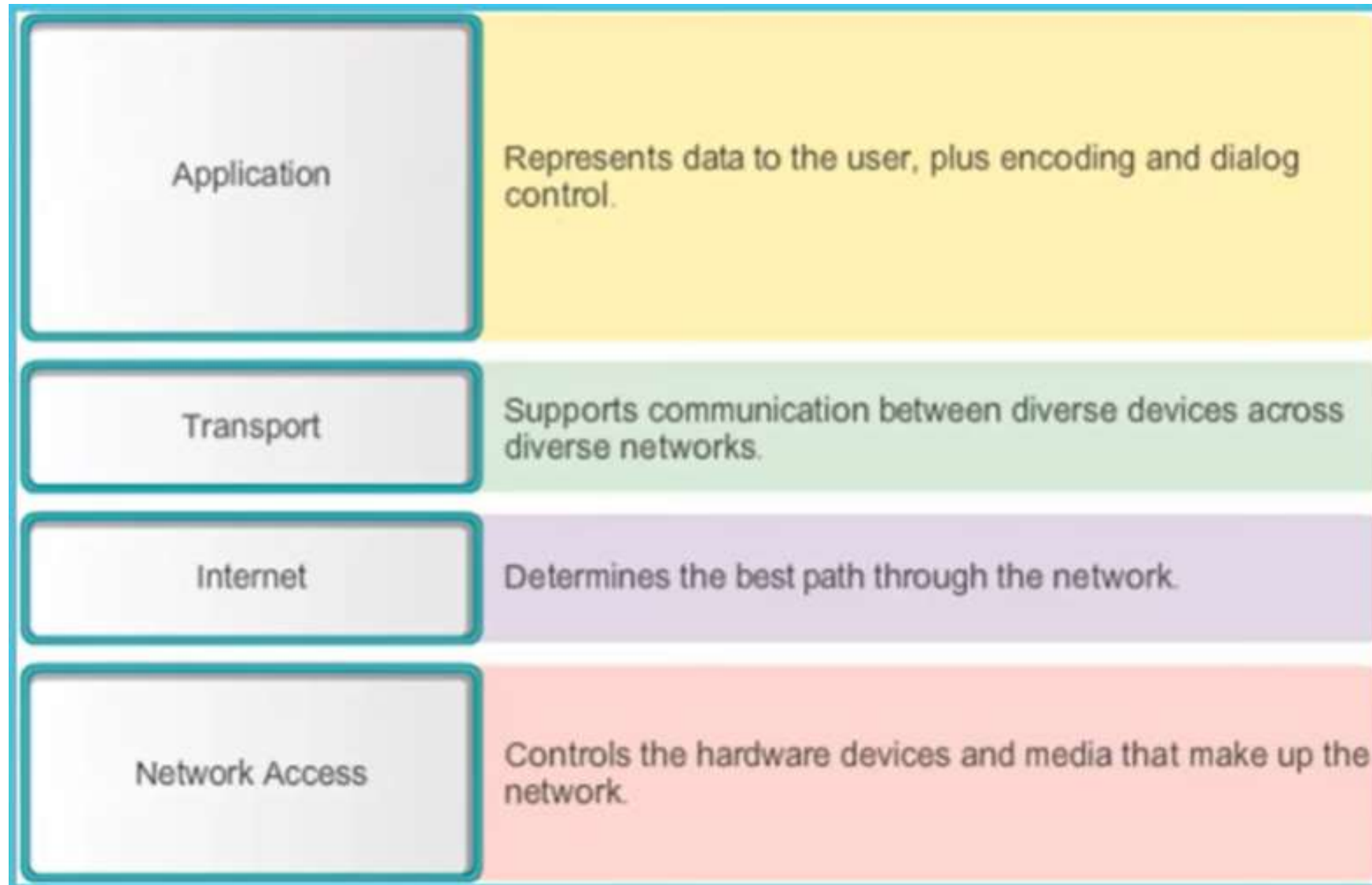
Physical Layer – Bits

TCP/IP Protocol Suite

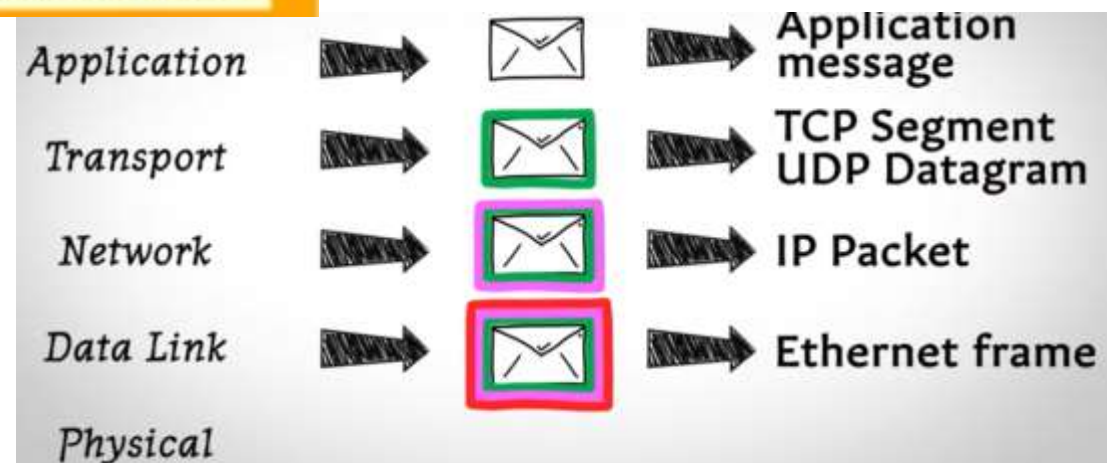
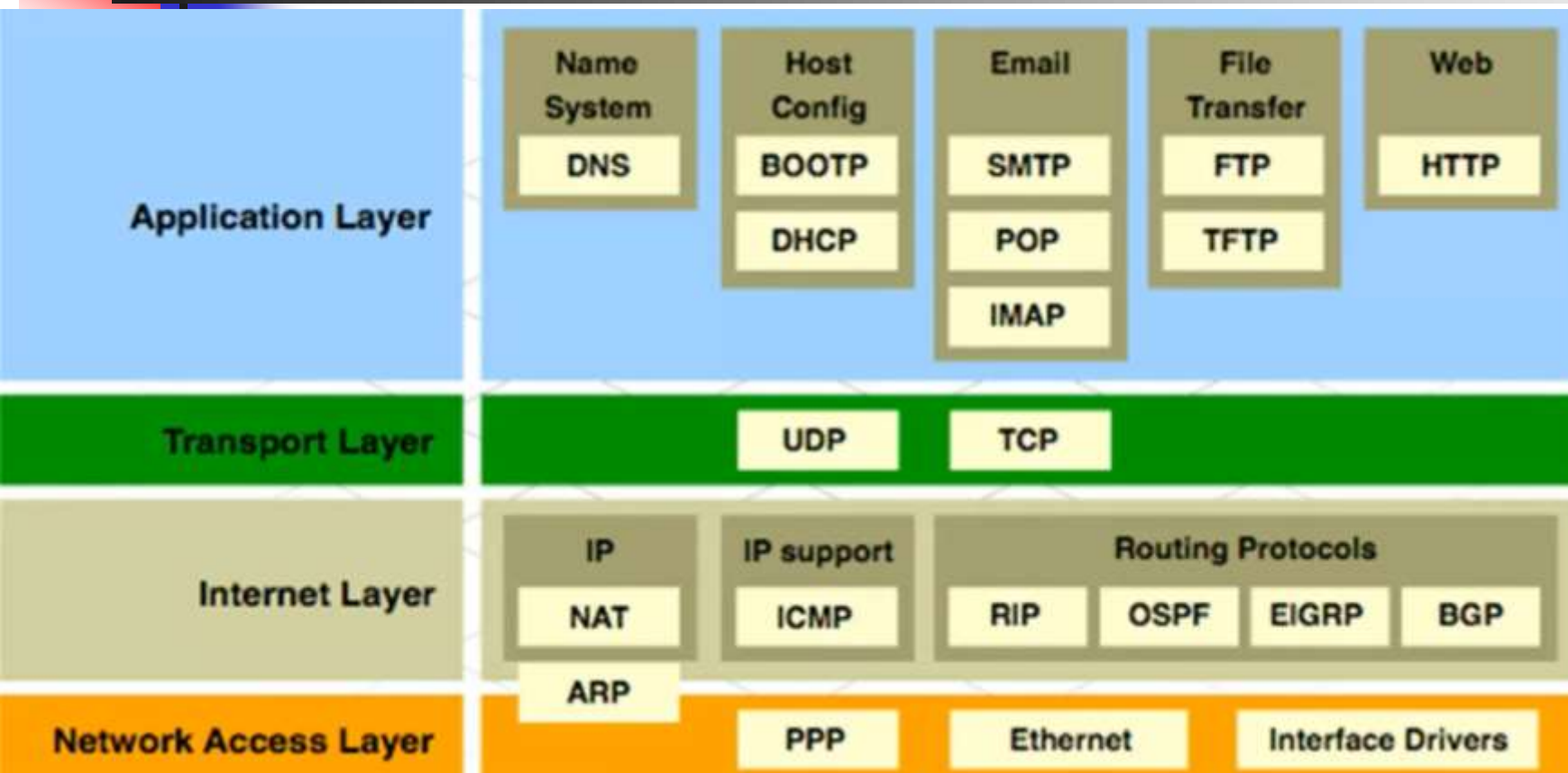
- ❖ TCP/IP protocol suite has four layers:
 - ✓ host-to-network,
 - ✓ internet,
 - ✓ transport, and
 - ✓ application.



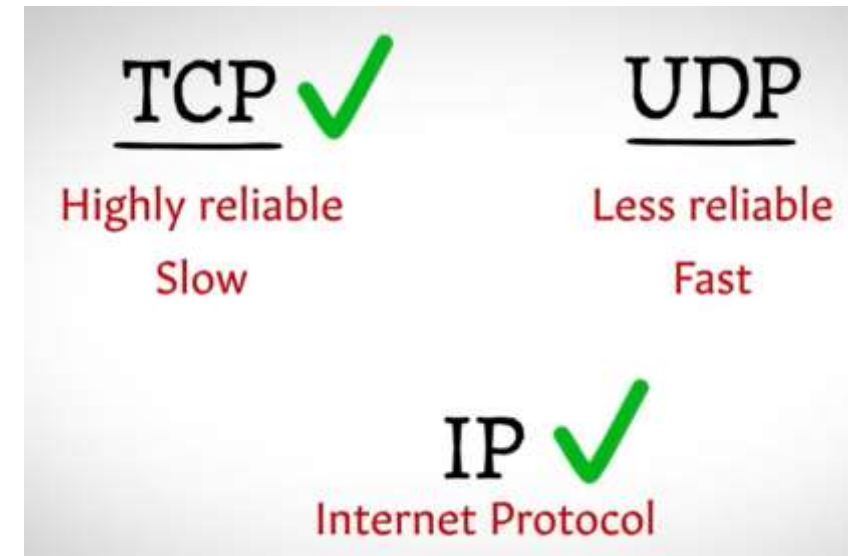
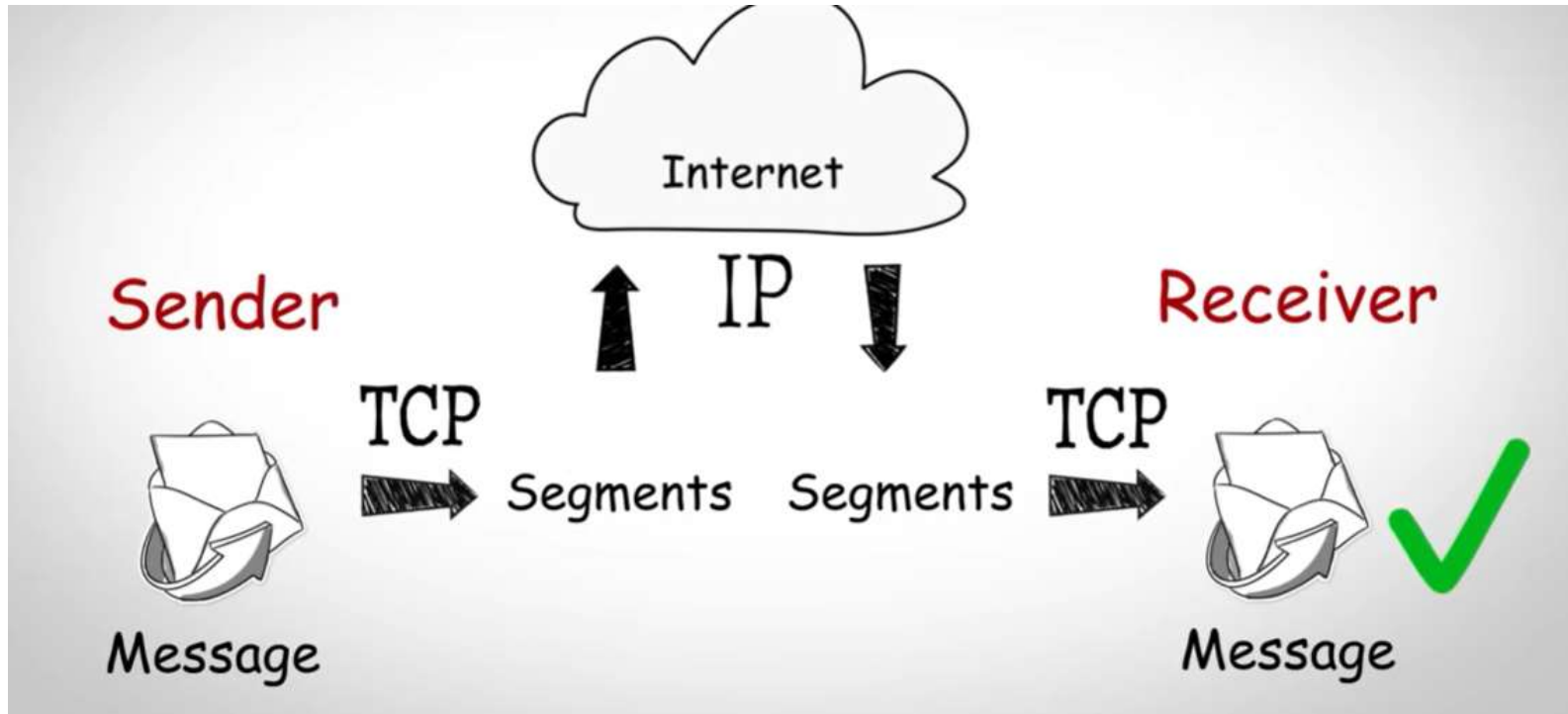
TCP/IP Protocol Suite



TCP/IP Protocol Suite



TCP/IP Protocol Suite





TCP/IP Protocol Suite

Physical layer protocol

Ethernet

Fast Ethernet
Gigabit Ethernet

Cable to be used

Twisted Pair Cable
Coaxial Cable
Fiber Optic Cable

Twisted Pair Cable
Fiber Optic Cable

TCP/IP Protocol Suite

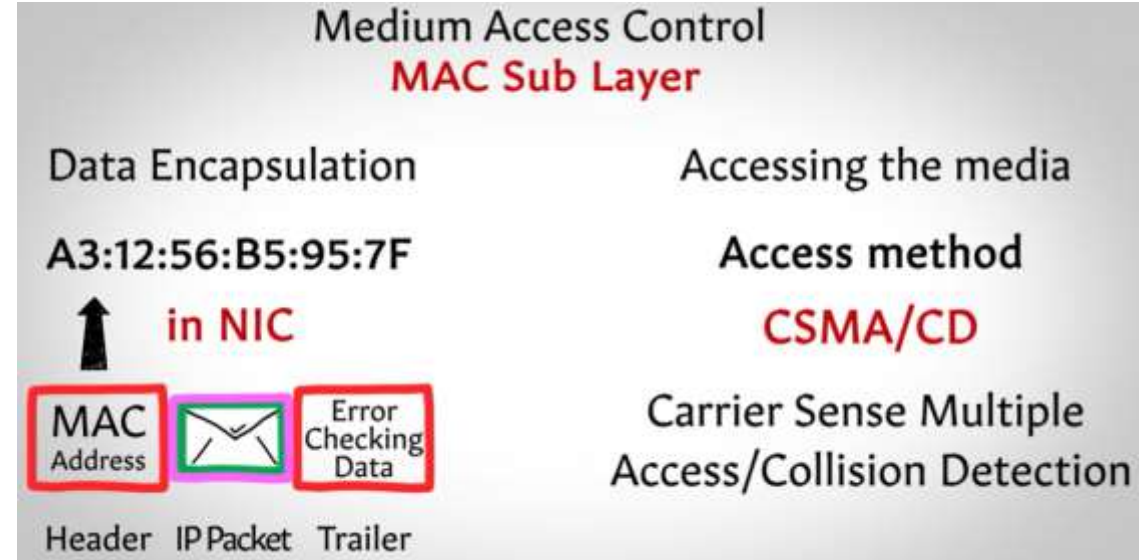
The Data Link Layer



Ethernet Frame

Medium Access Control
MAC Sub Layer

Logical Link Control
LLC Sub Layer



Logical Link Control
LLC Sublayer

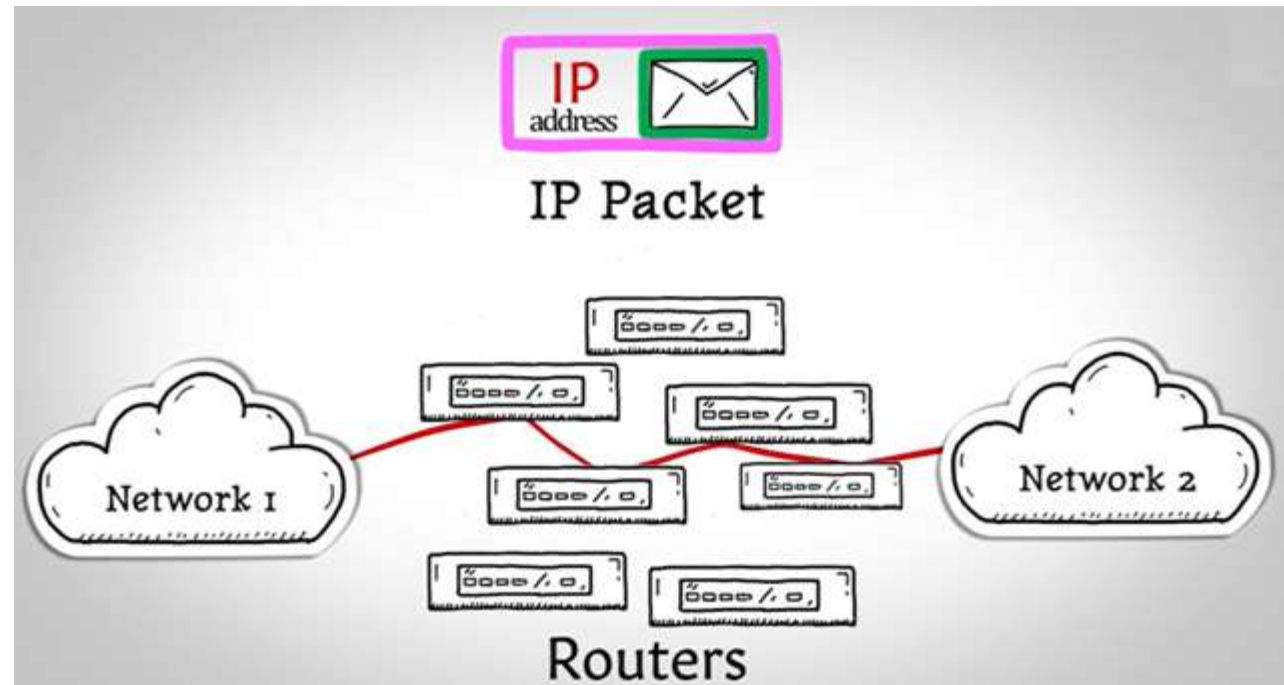
Flow Control
Error Control

TCP/IP Protocol Suite

❖ Network Layer:

Logical addressing

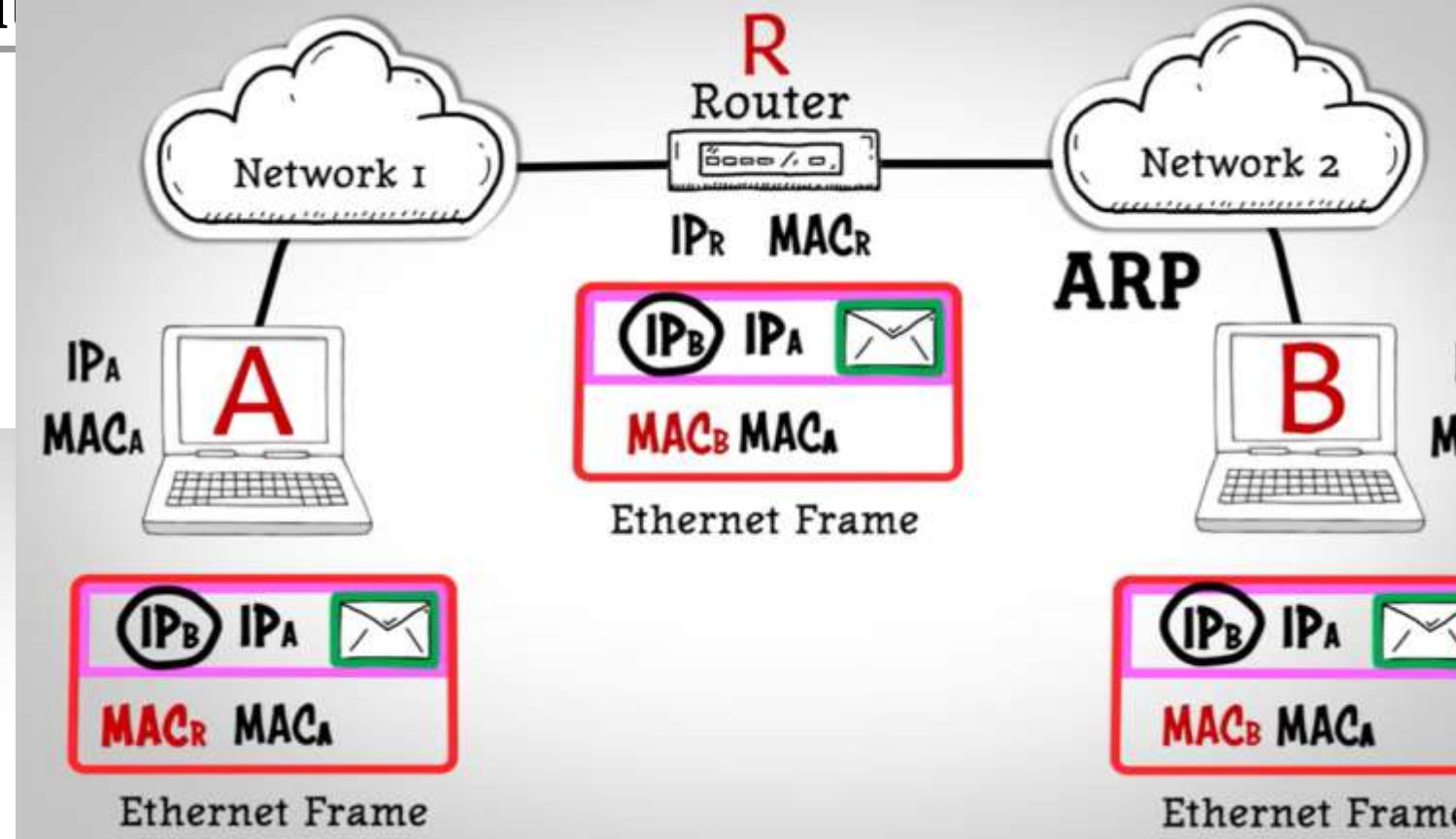
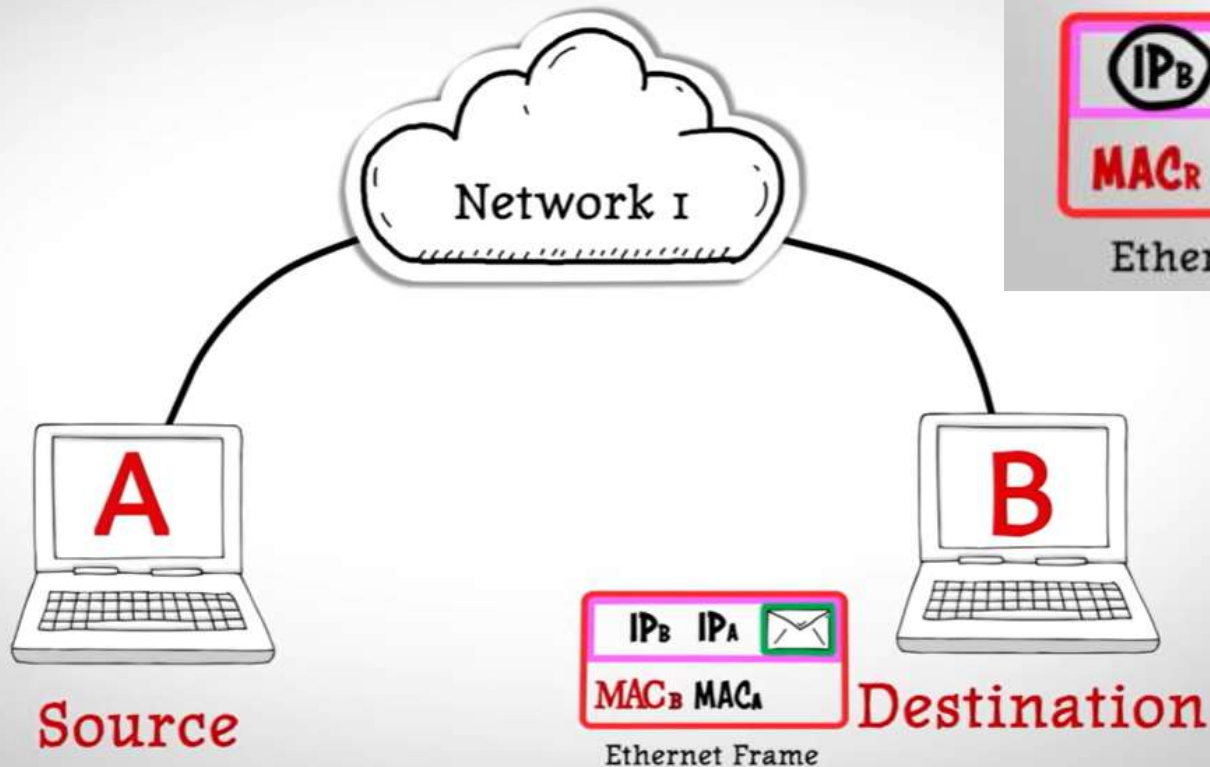
Routing



TCP/IP Protocol Suite

❖ Network Layer:

No routing is needed



❖ Network Layer:

At the network layer (or, more accurately, the internetwork layer), TCP/IP supports the Internetworking Protocol. IP, in turn, uses four supporting protocols: ARP, RARP, ICMP, and IGMP.

✓ Internetworking Protocol (IP)

The Internetworking Protocol (IP) is the transmission mechanism used by the TCP/IP protocols. It is an unreliable and connectionless protocol—a best-effort delivery service. The term best effort means that IP provides no error checking or tracking.

IP is unreliable

**It does not guarantee delivery
nor check for errors.**



TCP/IP Protocol Suite

✓ Address Resolution Protocol

ARP is used to find the physical address of the node when its Internet address is known.

✓ Reverse Address Resolution Protocol

The Reverse Address Resolution Protocol (RARP) allows a host to discover its Internet address when it knows only its physical address.

✓ Internet Control Message Protocol

The Internet Control Message Protocol (ICMP) is a mechanism used by hosts and gateways to send notification of datagram problems back to the sender.

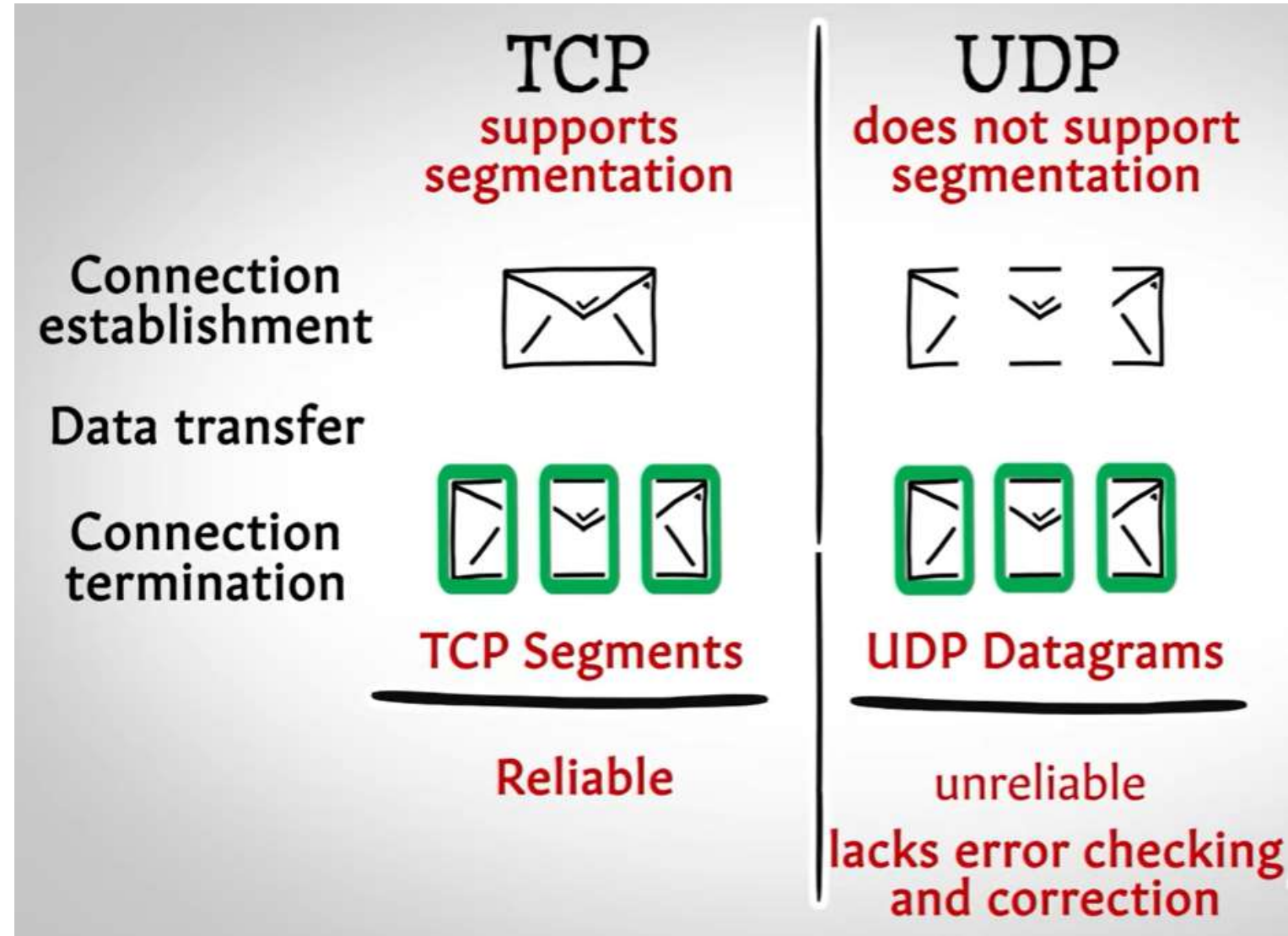
✓ Internet Group Message Protocol

The Internet Group Message Protocol (IGMP) is used to facilitate the simultaneous transmission of a message to a group of recipients.

TCP/IP Protocol Suite

❖ Transport Layer:

Traditionally the transport layer was represented in TCP/IP by two protocols: TCP and UDP. UDP and TCP are transport level protocols responsible for delivery of a message from a process (running program) to another process. A new transport layer protocol, SCTP, has been devised to meet the needs of some newer applications.

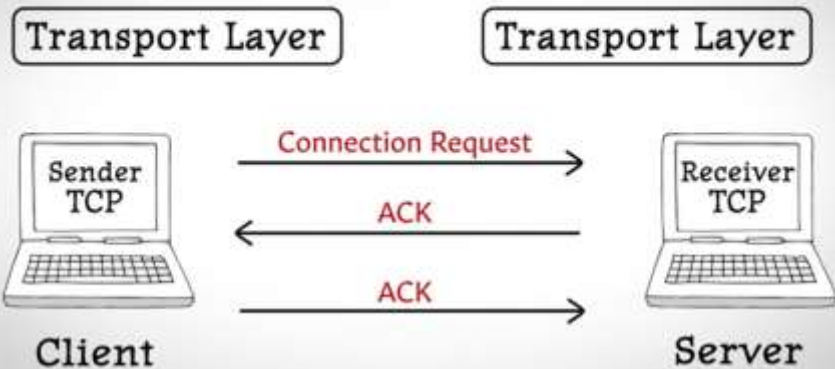


TCP/IP Protocol Suite

✓ Transmission Control Protocol

The Transmission Control Protocol (TCP) provides full transport-layer services to applications. TCP is a reliable stream transport protocol. The term stream, in this context, means connection-oriented: A connection must be established between both ends of a transmission before either can transmit data.

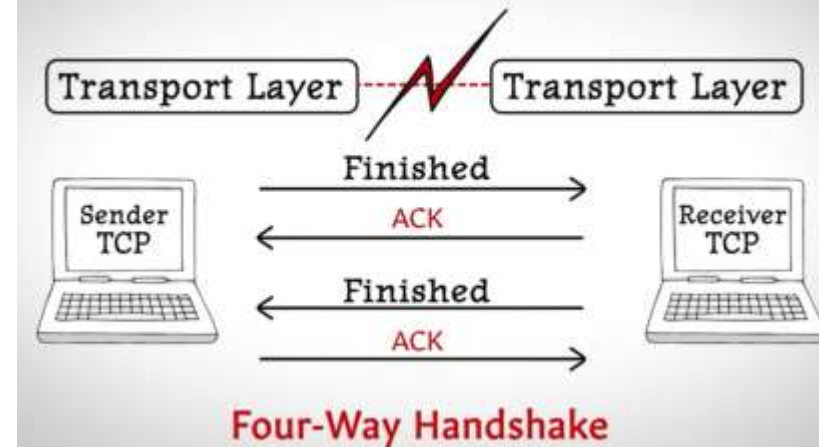
Connection establishment



Data Transfer

Error-free data transfer
Ordered-data transfer
Retransmission of lost data
Discarding duplicate packets

Connection termination





TCP/IP Protocol Suite

✓ User Datagram Protocol

The User Datagram Protocol (UDP) is the simpler of the two standard TCP/IP transport protocols. It is a process-to-process protocol that adds only port addresses, checksum error control, and length information to the data from the upper layer.

✓ Stream Control Transmission Protocol

The Stream Control Transmission Protocol (SCTP) provides support for newer applications such as voice over the Internet. It is a transport layer protocol that combines the best features of UDP and TCP.

❖ Application Layer

The application layer in TCP/IP is equivalent to the combined session, presentation, and application layers in the OSI model. Many protocols are defined at this layer.



Data Communication and Networking Syllabus

Unit 1: Introduction and Physical Layer

Overview of Computer Network, OSI and TCP/IP Reference Models, Guided and Unguided Transmission Media, Analog and Digital Communication, Encoding and Modulation, Nyquist Theorem, Shannon's capacity, Switching techniques- TDM, FDM.

Unit 2: Data Link Layer

Framing, Error detection and Error correction, Vertical Redundancy Check (VRC), Longitudinal Redundancy Check (LRC), Hamming Distance, Hamming Code, Cyclic Redundancy Check (CRC), Stop and Wait Protocol, Sliding Window Protocol, Go-Back-n ARQ, Selective-Reject ARQ, HDLC, Channel Allocation, ALOHA Systems, CSMA Protocols, Collision Free Protocols



Data Communication and Networking Syllabus

Unit 3: Network Layer

Introduction to the Network Layer, Network Layer Protocols IPv4 and IPv6: header structure and features, Comparison of IPv4 and IPv6, Transition strategies (Dual stack, Tunneling), Addressing in the Network Layer, Classful vs Classless addressing, Routing Fundamentals, Types of routing: static, dynamic, Routing protocol, Causes and effects of network congestion, Congestion control techniques: Leaky Bucket and Token Bucket, Network Address Translation (NAT)

Unit 4: Transport Layer

Process-To-Process delivery, user datagram, Transmission control protocol.



Data Communication and Networking Syllabus

Unit 5: Application Layer

Client-Server Model, Socket interface, A brief introduction to DNS, SMTP, FTP.

Unit 6: Network Security

Introduction to Network Security, Common threats: malware, phishing, and DoS attacks, Cryptography Basics, plaintext, ciphertext, encryption, decryption, Authentication and Authorization, passwords, two-factor authentication, and biometrics, Firewalls and VPNs



Data Communication and Networking Syllabus

Textbooks:

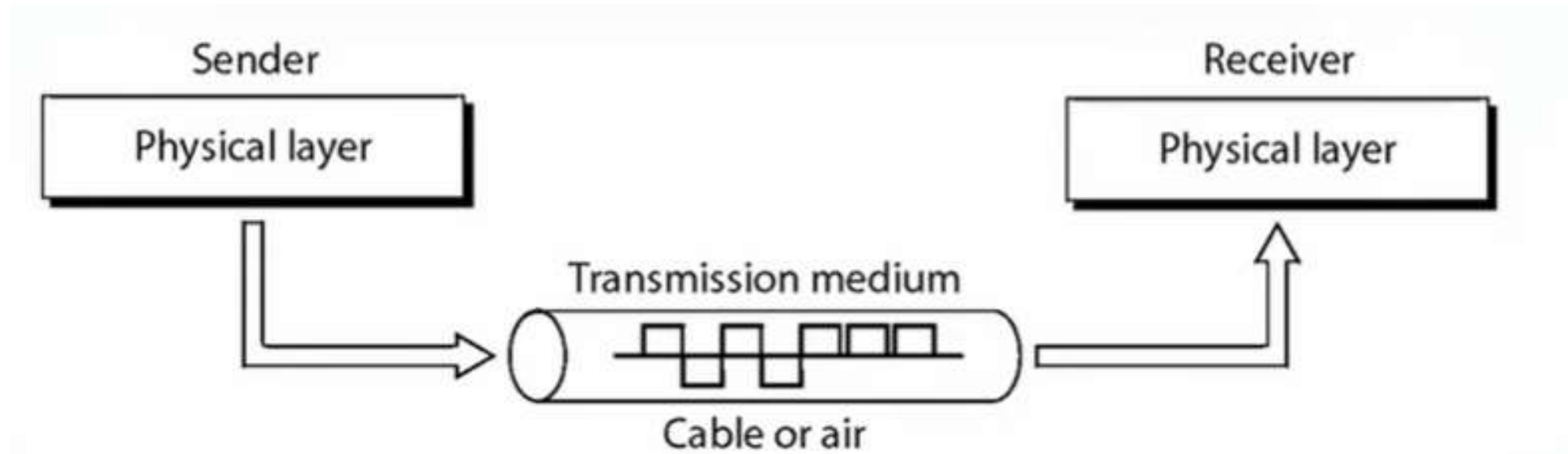
1. Data Communications & Networking, Behrouz A. Forouzan, Tata McGrawHill, 4th Edition, 2007
2. Data Communication and Computer Networks, Rajneesh Agarwal, Vikas Publication house, 2018, reprint
3. Computer Networks by Andrew Tanenbaum, Fourth Impression, Pearson Education India, 2010
4. Internetworking with TCP/IP, Vol. 3, Client-Server Programming and Applications by Douglas E. Comer, Prentice Hall Publisher, 2014

Unit 1: Introduction and Physical Layer

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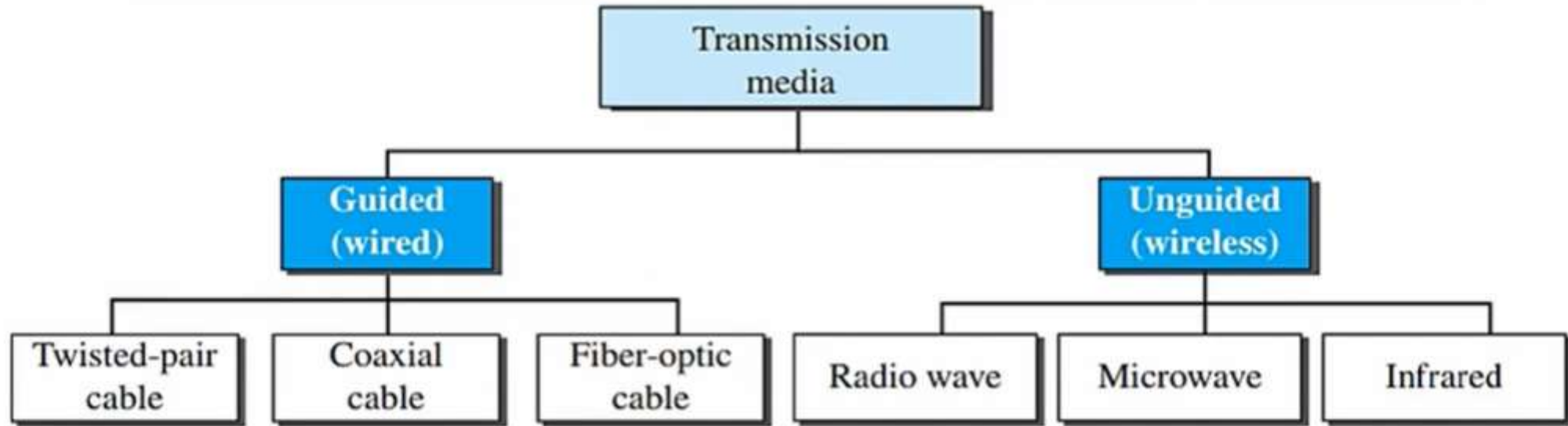
Guided and Unguided Transmission Media

- ❖ Transmission media is a communication channel that carries the information from the sender to the receiver
- ❖ Transmission medium is usually free space, metallic cable or fiber optic cable
- ❖ Transmission media work on the physical layer of OSI model



Guided and Unguided Transmission Media

Classification of Transmission media

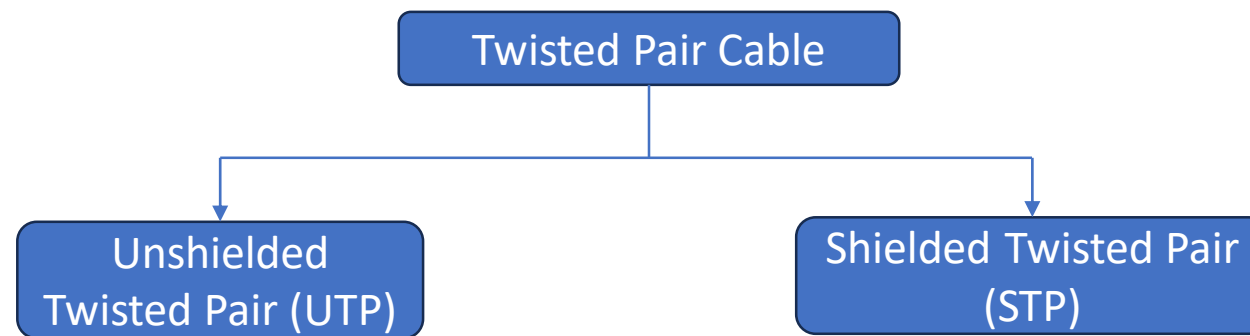
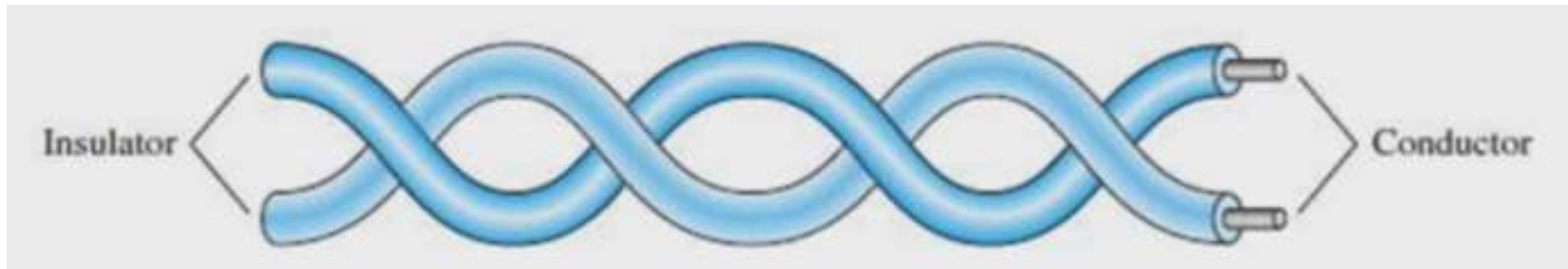


- ❖ Guided media has the physical media/cable for transmission of signal from one device to the other
- ❖ Unguided media uses the free space for the transmission of signal

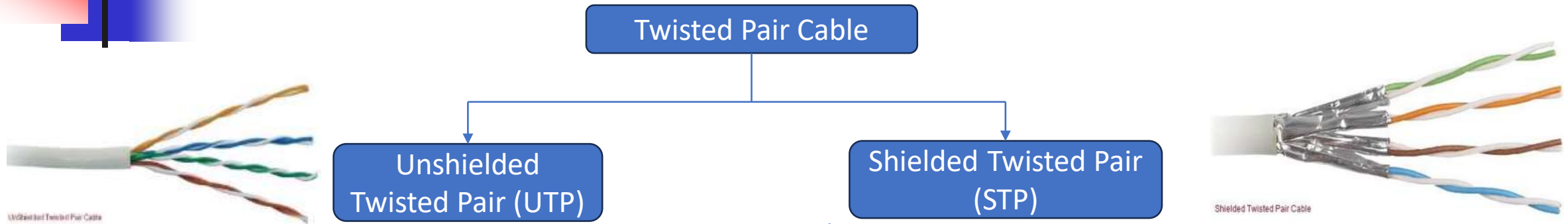
Guided and Unguided Transmission Media

Twisted Pair Cable

- ❖ A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern.
- ❖ Typically, a number of these pairs are bundled together into a cable by wrapping them in a tough protective sheath.
- ❖ Twisted pair (TP) used to transmit both analog and digital signal.



Guided and Unguided Transmission Media



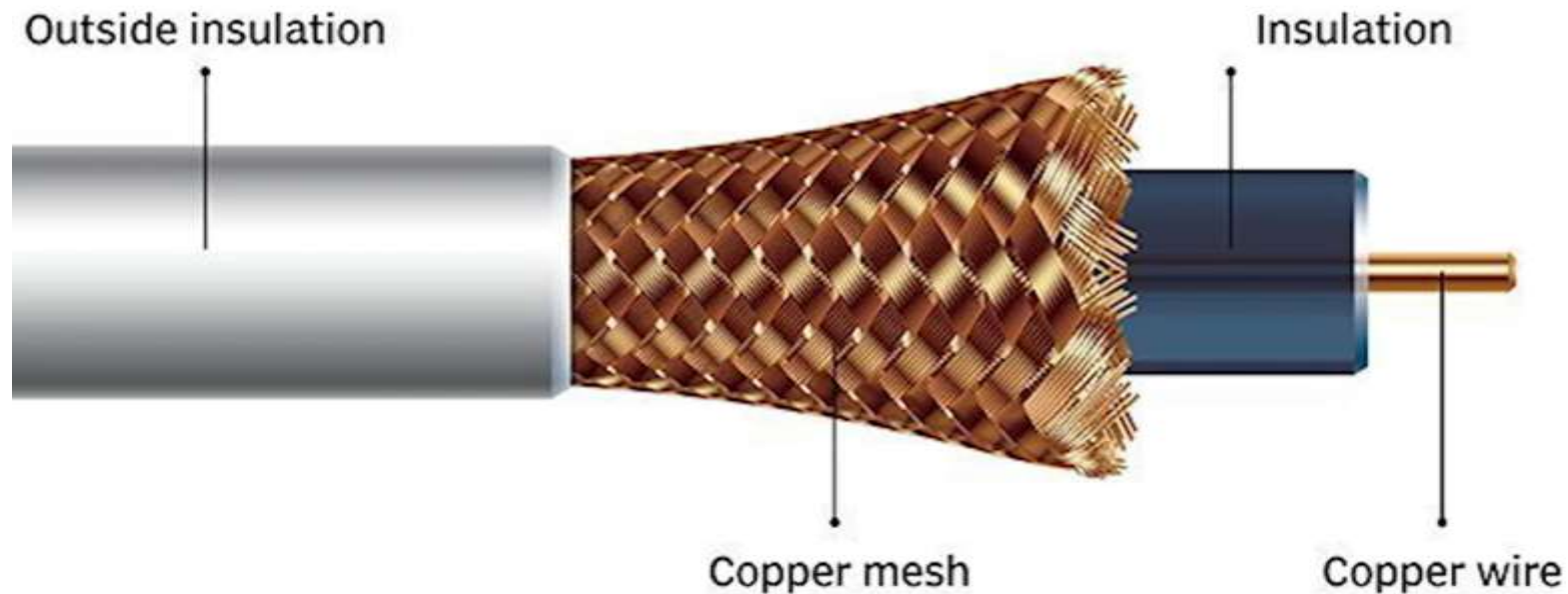
- ❖ UTP cables are the most common telecommunications medium to carry both voice and data transmission.
- ❖ UTP cables consist of 2 or 4 pairs of twisted cable.
- ❖ Installation is easy, cheap
- ❖ Range is less than 100m
- ❖ It is also used in Ethernet cables.
- ❖ Example: 10 base T, 100 base T and 10 base Tx

- ❖ STP cables have a shielding in usually of metal mesh (aluminum or polyester material) between the outer jacket and wire.
- ❖ The metal mesh around the insulated wires eliminates crosstalk
- ❖ Range is bit more compared to UTP

Guided and Unguided Transmission Media

Co-axial Cable

- ❖ The name coaxial is because it contains two conductors that are parallel to each other and share common axis.
- ❖ Inner conductor is made of copper which is surrounded by PVC insulation.
- ❖ The outer conductor is metal foil, mesh or both.





Guided and Unguided Transmission Media

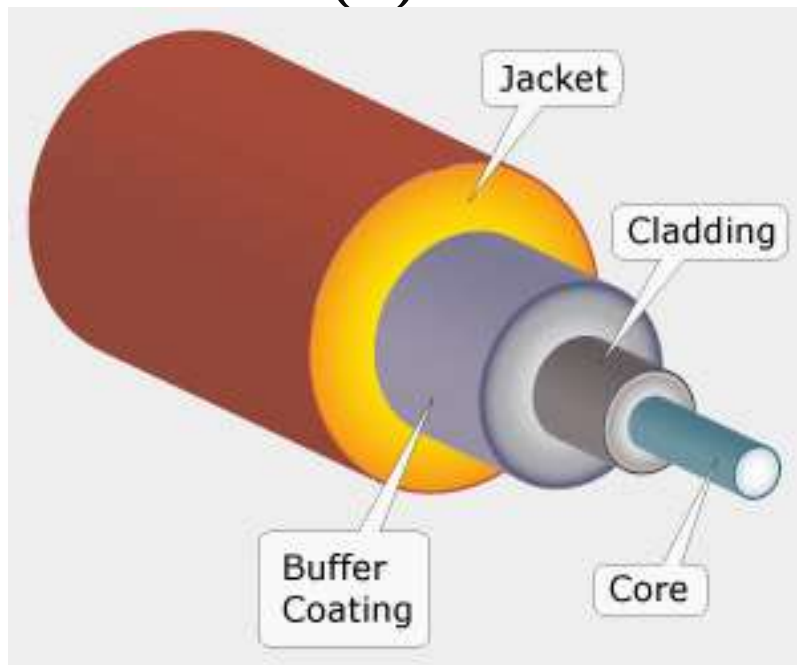
Co-axial Cable

- ❖ Outer metallic conductor is used as a shield against noise.
- ❖ The outer conductor is also encased in an insulating sheath.
- ❖ The outermost part is the plastic cover which protects the whole cable.
- ❖ Co-axial cable is much less liable to interference and cross talk than the twisted pair.
- ❖ It is widely used in cable TV and broadband connection
- ❖ Range is around 1km
- ❖ Cost and strength of the signal carried by the coaxial cable is higher when compared to twisted pair cable
- ❖ Example: 10 base 2, 100 base 5

Guided and Unguided Transmission Media

Fiber Optic Cable

- ❖ It is made of glass or plastic and transmits the signal in the form of light
- ❖ Optical fibers use the concept of reflection to guide the light through the channel
- ❖ A light pulse can be used to represent a one (1) bit. The absence of a pulse signal for zero (0).





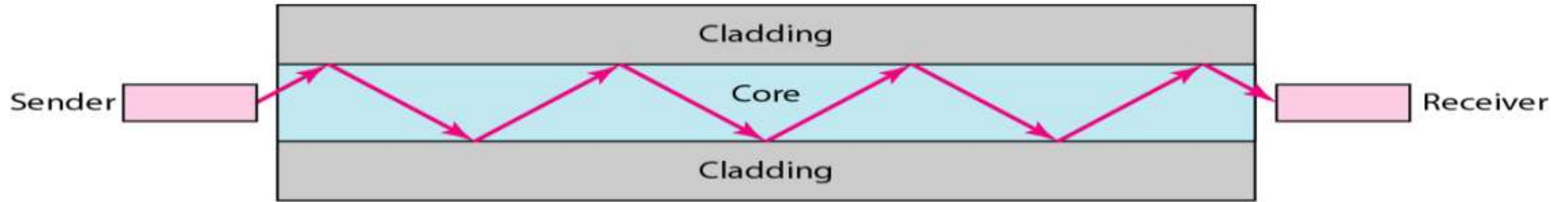
Guided and Unguided Transmission Media

Fiber Optic Cable

- ❖ An optical fiber has a cylindrical shape and consists of 3 concentric sections
 - ✓ Core - It's the innermost section, made of glass or plastic, and is surrounded by its own cladding. The core diameter is in the range of 8 to 50 μm .
 - ✓ Cladding - A glass or plastic coating that has optical properties different from those of the core, having a diameter of 125 μm . The cladding acts as a reflector to light that would otherwise escape the core.
 - ✓ Jacket - The outermost layer is the jacket. The jacket is composed of plastic or other material layer to protect against moisture, cut, crushing, and other environmental dangers.

Guided and Unguided Transmission Media

Fiber Optic Cable

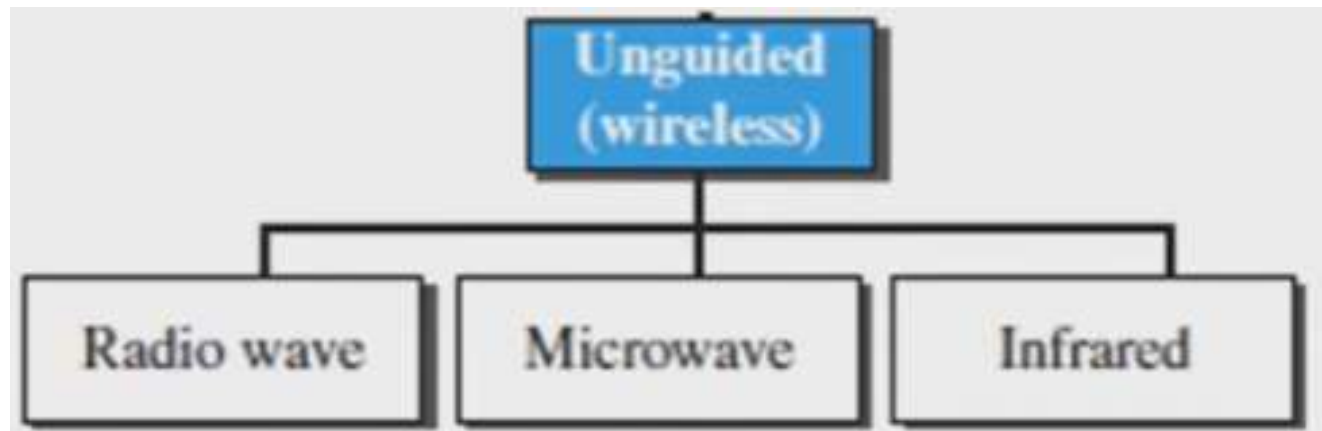


- ❖ It has very low noise interference
- ❖ It offers very high bandwidth
- ❖ Optic cables are used for long distance (in terms of few km)
- ❖ Earlier it was costly but now due to advanced fabrication technology, cost is going down
- ❖ Example: 1000 base Fx

Guided and Unguided Transmission Media

Unguided Transmission Media

- ❖ Transport the electromagnetic waves without using physical conductor
- ❖ Often referred as wireless communication
- ❖ Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them
- ❖ It is used for larger distances
- ❖ It is less secure



Guided and Unguided Transmission Media

Radio Waves

Frequency Range: 3kHz – 1GHz

Omnidirectional – Sending and Receiving antennas are not aligned



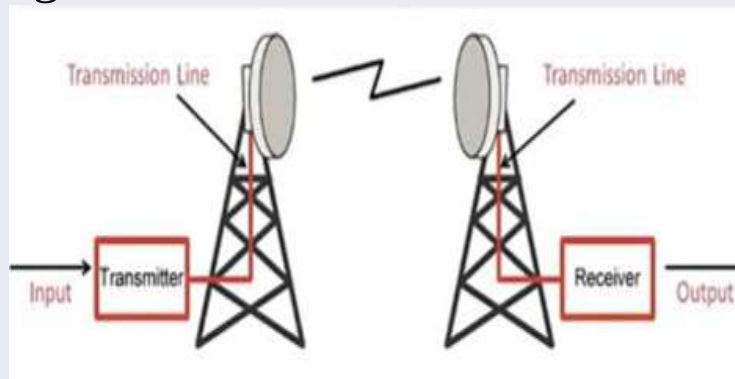
Higher data transmission rate

It is used broadcasting. radio and television

Microwave

Frequency Range: 1 GHz – 300 GHz

Unidirectional: The sending and receiving antennas needs to be aligned



Distance travelled is proportional to the height of the antenna

Unicast communication. cellular telephones, satellite networks, and wireless LANs.

Infrared

Frequency Range: 300GHz – 400 THz

Used for shorter distance using line-of-sight propagation

Data cannot pass through obstacles. TV remote, Wireless mouse

High Security